Supplementary material for
Specification Surfaces for Incremental Predictive Performance: Estimation, Uncertainty, and Multi-Log Evaluation

Sudhir Dixit[a]

[a]*Independent Researcher*

Abstract

This document holds the protocol and its amendments, the proofs, the constructions of the reporting objects, the correction register, the per-layer and per-fold results, the literature pilot, the diagnosis of the one simulated world where the bootstrap misbehaves, the worked example, the verification harness, the derivation of the coverage calibration, three results the article summarises in a paragraph each — the family-by-encoding crossing, the decision-rule comparison, and the calibration, unseen-category and rolling-origin diagnostics — and the tables the article does not print. The article is self-contained without it. Every number in both documents is generated from the same result files by the same script, so a quantity cannot differ between them.

Contents

Email address: sudhir.dixit1@gmail.com (Sudhir Dixit)

S1. Protocol, and the amendments to it

The corpus protocol is `PROTOCOL.md` and the practice pilot's is `AUDIT-PROTOCOL-2.md`; both are in the archive and both were committed before the first result they govern existed. This appendix lists what changed afterwards.

Why amendments are reported rather than folded in. A pre-registration that is quietly edited is worse than none, because it claims a discipline it does not have. Each amendment below names its date, the defect that forced it, and the result it moved.

Amendment 1 — a register's values must recur across time. Registered rule 3.3 admits any attribute reused across two cases on average. On BPI Challenge 2019 that admits the purchasing-document number, which is a document identifier and not a maintained entity: almost none of its values recur across the temporal split. The amendment adds a second, outcome-free condition — at least 50% of test cases must carry a value seen in training — and *both* selections are reported for every log.

Amendment 2 — "any timestamp" had to be implemented. Rule 3.1 excludes timestamps. Matching a fixed list of names did not implement that: it missed five differently-spelled date columns, two of which the registered rules then made the high-cost entity. A column is now a timestamp if its name looks like one or if more than 90% of its non-missing values parse as a date. Both tests read feature values only.

Amendment 3 — a relabelling of the activity is the activity. Some logs carry a second column that is a one-to-one recoding of `concept:name`. Admitting it as a feature admits the activity sequence, which is not available at case creation.

Amendment 4 — coarsenings of the register get their own role. Type, subtype and service are deterministic functions of the item. Sweeping them into the intake block makes the intake block partly a register, and the increment then measures the residual rather than the field. They are a separate role and S4 uses it.

Amendment 5 — the caused-by family is an outcome. A caused-by attribution is the output of diagnosis, not an intake observation. Without this the registered rules made the caused-by configuration item the high-cost entity on the flagship log.

Amendment 6 — the pilot frame's topical filter was nulled. The frame's keyword filter was applied to titles and abstracts, which is a filter on how a paper describes itself rather than on what it does. `results/s06_frame.csv` reports what nulling it does to the estimates.

What the protocol fixed, before any of the results it governs. The design space of Section 3; the reference specification; the exclusion of the intercept-only rung from every admissible set; the six register-quality mechanisms and their levels; the five planned contrasts and the Holm procedure; the two families over which simultaneous bands are taken; and the two-stratum design of the practice pilot, including the decision to adjudicate a sample of the papers the screen *rejected*. The commit that fixed them adds `scripts/spec.py` and `PLAN-ROUND19.md` and no result file.

S1.1. Related work, strand by strand

Four further strands bear on the argument. *Many-analysts studies* put one question and one dataset to independent teams and observe the spread directly [29, 3]: they establish that analytic variability is large without supplying an estimand for it, which is what the surface is. *Predictive multiplicity* and Rashomon sets study near-equally accurate models that disagree on individuals [21, 27] — multiplicity at the level of a fitted model, where this paper's is at the level of the design that produced it. *Benchmark-variance work in predictive process monitoring* reports how far results move with the pipeline and the protocol [26], which is the evidence this paper's decomposition partitions. And the *sensitivity-analysis* tradition of bounding an unmeasured assumption [36] is the form Section 7.3's leakage tipping point takes. On the estate side, the accuracy of a configuration management database is itself a studied quantity [33, 23], and the layered service models practitioners maintain are why Section 7.4 asks which *layer* of a register pays. Supplement S1.1 sets out each strand and what this paper takes from it.

Variable importance and incremental value. That a feature's apparent importance depends on what else is in the model is the standard motivation for conditional and permutation-based importance [32, 11], for Shapley-value attributions [19, 8], and for nonparametric formulations of importance as a population quantity [43], which average over subsets precisely because a single conditioning set is arbitrary. The clinical-prediction literature has made the same point about the *incremental* value of a marker for two decades: the increase in the area under the ROC curve from adding a predictor depends on how good the existing model is [25, 7], is a poor guide to clinical usefulness [7, 38], and is routinely misread [16]. What that literature establishes, and what we take as given, is that the baseline and the metric are choices. What it does not supply is an object that reports the whole dependence, an estimator for it, or inference over it.

Decision-analytic evaluation. Decision curve analysis [40, 41] makes the operating point explicit by reading net benefit across an exchange rate, and is the reason the operating point is an axis here rather than a footnote. Guidance on reading such a curve is well developed [39]; what is less settled is what a claim about a *band* of thresholds requires, which is why the bands in Section 8 are simultaneous rather than pointwise.

Specification sensitivity. That analytic conclusions move with defensible analytic choices is the subject of multiverse analysis [31], specification-curve analysis [30], vibration of effects [24] and, in machine learning evaluation, of work on underspecification [9] and on the fragility of benchmark rankings [2]. Our design space and our figures are recognisably in that tradition. What this paper adds to it is a decomposition of the surface's variance into axis contributions, simultaneous inference over the surface rather than a distribution of point estimates, region labels with an explicit decision rule, and a regret benchmark against the conventional report. Specification-curve analysis, as usually practised, displays the curve and applies a permutation test to its median; it does not tell a reader which sub-region of the design space supports a sign.

Data valuation. Shapley-based data valuation [12, 13] values *instances* against a fixed learner and metric; the axes we vary are exactly the ones held fixed there. The two are complementary: a data value is itself a point on a specification surface.

Which axis a study varies. Some studies do vary one axis deliberately: the metric axis in effort-aware defect prediction [4], the encoding axis in remaining-time prediction [18], the inter-case feature block in predictive process monitoring [28]. What is missing is an object that holds several axes at once and says what their joint variation implies for a sign.

Prediction on incident event logs. The empirical setting is predictive process monitoring, surveyed and benchmarked across real logs [34, 37] on data of the kind process mining formalises [1]. We predict at case creation from creation-time attributes with no prefix, which is tabular classification on a process log rather than prefix-based monitoring, and that difference bounds what the corpus result transfers to. The benchmark protocols in that literature fix a learner, an encoding and a metric per experiment, and this paper measures the sensitivity of a conclusion to *those* fixings. It does not measure sensitivity to **prefix length, prefix bucketing or sequence encoding**, which are the axes that literature varies most, because none of them exists at a single prediction point; Section 11 carries the transfer as an assumption rather than a result. Configuration management databases and their data quality are treated in the IT service management literature as an operational problem [20]; we are not aware of an evaluation of one against a named prediction task with the design space held open.

Data quality as an axis. The effect of data quality on model performance has been studied directly on tabular data [23] and, in process mining, for low-quality activity labels [6]. Those studies vary quality with the feature set fixed; we vary quality *as one axis of the same surface* on which the feature set and the encoding are others, which is what makes the interaction between them visible — Section 7.3 reports a quality defect that is exactly invisible under one encoder and not under another. Leakage, which is availability misspecified, has its own literature [15, 14, 42], and the decision time of Section 7.2 makes it a declared choice rather than a mistake.

S1.2. The measure on the axes, and the two axes that are not analysis choices

On the measure. A design space is a set; a decomposition, a weighted mean or a robustness index needs a *distribution* on it, and "uniform over the cells I enumerated" is not neutral — it moves if a level is duplicated, if five folds are written as five levels rather than one, or if a grid is refined. This paper separates the weight on an axis from the weights within it, declares three product measures and an envelope over a fourth (Section 4.5), and reports every weighted quantity under all of them.

Two of these axes deserve comment because they are usually not treated as design choices at all.

The **decision time** τ fixes which fields exist when the prediction is made. It is not the same as the split rule and it is not a data-cleaning detail: moving τ later in a process admits fields written in between, and Section 7 is a case in which that single move takes a field's increment from resolvably positive to indistinguishable from zero.

The **register quality** q acts on f rather than on the comparison. A field that is a lookup into a maintained register — a configuration item, a customer master record, a product catalogue — inherits that register's completeness and accuracy. We implement six mechanisms besides the clean field, each parameterised by the training half alone: population loss with the long tail absent, with the core absent, and at random; identity error, in which a share of rows carry a wrong value drawn from the training alphabet; reconciliation failure, in which a share of identities exist twice; and staleness, in which an identity is present only if the register had already seen it by the split point. The first three model coverage, the next two accuracy, the last discovery; the taxonomy follows the event-log imperfection patterns of Suriadi et al. [33] where they apply.

Real registers do not fail at random, so three *further* mechanisms make the missingness depend on something: on *time*, where a row's probability of carrying an identity rises with its position in the log; on *content*, where coverage depends on the item's recorded type through a map estimated on the training half; and on a *propensity* built from the intake block, which makes missingness correlated with the outcome without ever reading a test outcome. Every mechanism is swept over 11 severities and every stochastic one repeated over 20 seeds, and each is verified by execution to be a function of the training half alone — perturbing the test half and re-degrading must leave the training half's degraded values identical, which `results/s01_trainonly.csv` records over 122 checks.

S1.3. Targets, splits and the exclusion codes

S1.4. Targets, splits and exclusions

Two targets are defined for every log: *handover*, whether more than one resource group touches the case, and *duration*, whether the case runs longer than the training half's median. Cases are ordered by their first timestamp and split 70/30; the rolling-origin scheme adds five expanding-window folds. Fields that are determined only at closure are excluded by name, which is the standard leakage discipline [15, 14] and, for event logs specifically, the discipline of Weytjens and De Weerdt [42]. A log–target pair is excluded only for a registered reason — too few cases, no variation in the test half, a prevalence outside $[0.05, 0.95]$, or the absence of one of the three roles — and every exclusion is reported with its code in Table S12, which accounts for every downloaded log that is not among the admitted pairs — including the held-out set, 1 of the checksummed files, downloaded for a separate test and never offered to these rules at all.

Every admitted pair is analysed and reported. Running the full surface only on the logs whose *reduction* is resolvable would condition the analysis on the outcome. The surface is computed on all 19 admitted pairs through one code path, and the pairs on which the register is worth nothing are in the same tables as the rest.

S1.5. The admission margin at the upper edge

The pair carrying the most cells sits closest to the admission edge. BPIC14/handover has the highest prevalence of any admitted pair, 0.927, against the window's upper bound of 0.95; the nearest pair the headroom rule turns away, BPIC15_5/handover, is at 0.959, and 5 pairs in all are excluded above the bound. That pair is also the largest single contributor to the corpus, carrying 4,800 of the 29,040 admissible scalar cells. A rule with a sharp edge decides some case narrowly, and this is that case: the corpus's biggest component is admitted on a margin narrower than the gap to the pairs excluded beside it. Table S17 reports every headline statistic on the ITSM-like subset as well, which is the population that does not turn on this decision.

S2. Proof details and constructions

Remark S1 (What survives a declared family of recalibrations)**.** *For a family Ψ of strictly increasing maps applied simultaneously to all four arms of a configuration, the reduction is Ψ-invariant exactly when the metric is* affine *under Ψ. Rank-basedness is sufficient and not necessary, and the gap is family-relative: the class-mean gap — the difference in mean score between the two outcome classes, which is not rank-based — has a reduction invariant to* 1.9×10^{-14} *inside the affine family and shifting by up to 0.638 over 5 non-affine ones. What this buys is that a search becomes a proof: one map and one configuration violating the invariance rules it out everywhere, and 18 such certificates are computed, the smallest moving the reduction by 0.241,* *against 2,400 checks in which the two rank-based instruments hold to exactly zero.*

Remark 1: the two-point algebra in full

Let a scoring rule take one of two values, with a of the P positives and b of the N negatives in the high block; write $\alpha = a/P$, $\beta = b/N$, $\rho = N/P$ and $\pi = P/(P+N) = 1/(1+\rho)$. Two exact identities follow directly from the definitions in Section 3.5:

$$\mathrm{AUC} = \tfrac{1}{2} + \tfrac{\alpha-\beta}{2}, \qquad \mathrm{AP} = \alpha\,\frac{\alpha}{\alpha+\beta\rho} + (1-\alpha)\pi.$$

The first is the Mann–Whitney statistic with ties at 1/2; the second is the step-wise sum over the two operating points the rule admits. Hence

$$\mathrm{AP} - \pi \;=\; \alpha\left(\frac{\alpha}{\alpha+\beta\rho} - \pi\right).$$

Fix $d = \alpha - \beta > 0$. Every rule on the line $\beta = \alpha - d$ has the same AUC increment $d/2$ over the fully tied baseline, so any two of them give $R_{\mathrm{AUC}} = 0$ exactly. Along that line the AP increment runs from $d(1-\pi)$ at $\alpha = d$, $\beta = 0$ down to $\dfrac{d\rho}{\big(1+(1-d)\rho\big)(1+\rho)}$ at $\alpha = 1$, a ratio of about $1+\rho$. How far apart the two reductions can be pushed therefore grows with the imbalance, which is a property of the construction and not a hidden assumption: refuting the existence of φ needs only that the two be *different*, and at the modest $\rho = 3$ this paper uses they differ by 0.417.

Remark S1: the cancelling configuration, and what a certificate is

Section 3.5 notes a configuration on which a non-rank-based metric's increment moves under ψ while the ratio does not. The construction is immediate once stated: take $\sigma_{B_1} = \sigma_{B_0}$ and $\sigma_{B_1\cup f} = \sigma_{B_0\cup f}$. Then $V(B_1) = V(B_0)$ for every metric and every ψ, so $R = 0$ identically, while V itself moves under ψ whenever m is not rank-based. The configuration is degenerate — the two baselines induce the same scores — and that is why the proposition quantifies over configurations rather than asserting a configuration-by-configuration change.

A *certificate* for an instrument m and a recalibration ψ is a triple of score vectors (a, b, c) on one labelled test set for which

$$\frac{m(b) - m(a)}{m(c) - m(a)} \neq \frac{m(\psi b) - m(\psi a)}{m(\psi c) - m(\psi a)}.$$

By Remark S1 the existence of one such triple establishes that ψ does not act affinely on m, and therefore that R_m is not invariant — a conclusion about *every* configuration drawn from a finite object. The certificates are searched for by maximising the gap over random triples; `results/s29_certificates.csv` records, for every instrument and every recalibration family, the triple found, the two ratios, and the resulting shift in R. The search reports failure where it fails, and a failure is not evidence of invariance.

A withdrawn computational result

A previous draft reported, as a labelled computational result, a search over a family of synthetic estates for witnesses that no design axis determines another. It is withdrawn and does not appear in this manuscript. Its answer moved between zero and five of twelve ordered pairs as an arbitrary equality tolerance was swept; failing to find a witness is not evidence that one axis determines another; and finding witnesses in a finite synthetic family is not a non-redundancy theorem. Nothing in the manuscript depended on it, which is the clearest argument for removing it rather than repairing it. This is the correction register in the archive.

Remark 1: the construction

Proof (construction). Take P positives and N negatives and let every score take one of two values, so a scoring rule is described by the pair (a, b) counting positives and negatives in the high block; write $\alpha = a/P$, $\beta = b/N$, $\rho = N/P$. Both baselines are the fully tied rule $\alpha = \beta = 1$, which has AUC $= 1/2$ and AP $= \pi$, the prevalence. Then exactly

$$\text{AUC} - \tfrac{1}{2} = \tfrac{\alpha - \beta}{2}, \qquad \text{AP} - \pi = \alpha \left(\frac{\alpha}{\alpha + \beta\rho} - \pi \right).$$

Fixing $\alpha - \beta$ equal on the two legs makes the AUC increments equal, hence $R_{\text{AUC}} = 0$ exactly — in integers, not to rounding — while sliding α along the admissible range moves the AP increments and sweeps R_{AP}. With

$P = 200$, $N = 600$ and therefore $\rho = 3$, the two configurations in `results/s29_prop1.csv` — (α, β) of $(0.8, 0.4)$ and $(0.6, 0.2)$ in one, $(0.9, 0.5)$ and $(0.5, 0.1)$ in the other — have identical R_{AUC} to 2.2×10^{-16} and R_{AP} of -0.250 against -0.667, differing by 0.417. A single φ cannot take one input to two outputs.

The construction is only a proof if those two conditions hold, so `s29_props.py` asserts both: it refuses to write its output if the two configurations fail to share R_{AUC} or fail to separate R_{AP}. A far more imbalanced problem — $\rho = 500$, which this construction previously used — drives the gap close to one, and the modest ρ here shows the separation does not depend on extreme imbalance. □

Remark S1: the proof

The equivalence's harder direction needs one assumption, which round twenty-six moved here from the article with the propositions themselves.

Assumption S1 (Richness)**.** Every quadruple of score vectors on the fixed labelled test set is realisable as a configuration. This is the assumption that no restriction is placed on the model class; it is what makes "for every configuration" quantify over a set with more than three elements, and it fails only if the analyst commits in advance to a family of scoring rules narrow enough to constrain the four arms jointly. It is used only for (i) $\Rightarrow$ (ii); the direction that matters in practice needs none.

Proof. (ii) $\Rightarrow$ (i). Each of $V_m(B_0)$ and $V_m(B_1)$ is a difference of two values of m, so an affine change of scale multiplies both by $A(\psi)$ and the additive constant cancels; their ratio, and hence R_m, is unchanged wherever the denominator is nonzero.

(i) $\Rightarrow$ (ii). Fix ψ and the test set, and write $m(\sigma)$ for $m(\sigma, y)$. For score vectors a, b, c, Assumption S1 admits the configuration $\sigma_{B_0} = a$, $\sigma_{B_0 \cup f} = c$, $\sigma_{B_1} = a$, $\sigma_{B_1 \cup f} = b$, for which $R_m = 1 - \big(m(b) - m(a)\big)/\big(m(c) - m(a)\big)$. Invariance says

$$\frac{m(b) - m(a)}{m(c) - m(a)} = \frac{m(\psi b) - m(\psi a)}{m(\psi c) - m(\psi a)} \tag{1}$$

for every admissible triple. First, the map $T : m(\sigma) \mapsto m(\psi\sigma)$ is well defined: if $m(b) = m(b')$ then applying (1) to (a, b, c) and to (a, b', c) with the same a, c gives $m(\psi b) = m(\psi b')$. Second, fix a and c and put $k = \big(T(m(c)) - T(m(a))\big)/\big(m(c) - m(a)\big)$; then (1) reads $T(u) = T(m(a)) + k\,(u - m(a))$ for every u in the range of m, which is affine with slope $A(\psi) = k$ and intercept $B(\psi) = T(m(a)) - k\,m(a)$. The slope is nonzero because otherwise the transformed denominator vanishes and R_m is undefined rather than invariant. □

Proposition S1: the 3×3 *construction*

Proof (construction). Two levels per axis will not do, and the reason is worth stating: with two levels the two main effects are $|P + Q|$ and $|P - Q|$, where P and Q are the two within-axis differences, so their ordering is decided by $\operatorname{sign}(PQ)$, and a strictly increasing ϕ preserves the sign of each difference. At two levels the ordering is therefore *invariant* and the proposition is false. From three levels a conditional mean averages more than two cells and a convex ϕ reweights them unequally. The 3×3 design in `results/s07_prop3.csv` has $S_A = 0.421$ against $S_B = 0.284$ in its own units and $S_A = 0.232$ against $S_B = 0.390$ under $\phi = \sqrt{\cdot}$, so the leader changes. □

Why the second proposition is stated relative to a family

It is stated *relative to a declared family of recalibrations*, which is both what its proof reaches and what an analyst can act on: nobody recalibrates by an arbitrary monotone map, and a reader who has declared that they will consider Platt scaling, or isotonic regression, or any affine rescaling, is asking about that family and not about all of them.

Richness is used only in the direction (i) $\Rightarrow$ (ii), so under a model class too narrow to realise the separating configuration an instrument can be invariant on *that class* without acting affinely; the direction that matters for practice, (ii) $\Rightarrow$ (i), needs no richness at all. A reader working inside a restricted family should read the proposition as a sufficient condition and establish non-invariance on their own class by exhibiting a configuration in it, which is what the certificates do.

S3. Construction of the inferential objects

Section 4 states what each of the reporting objects is and why it is that and not something else. This appendix carries the constructions themselves, so that the section can be read for the argument and this one for the arithmetic. What is here is the detail a reader would need to reimplement the objects without reading the code.

The multiplier bootstrap for the max-t critical value

Let $\mathcal{F}$ be the declared family, $V_c^{(b)}$ the bth bootstrap draw at cell c, $\bar{V}_c$ the draw mean and $\hat{\sigma}_c$ the per-cell bootstrap standard error. Standardise the draw matrix,

$$Z_{b,c} \;=\; \frac{V_c^{(b)} - \bar{V}_c}{\hat{\sigma}_c}, \qquad b = 1, \ldots, B, \quad c \in \mathcal{F},$$

and let $e_1, \ldots, e_B$ be independent standard normals drawn afresh for each replicate. The *multiplier* statistic is

$$T^{(\mathrm{mult})} \;=\; \max_{c \in \mathcal{F}} \Big| B^{-1/2} \textstyle\sum_b e_b Z_{b,c} \Big|.$$

Conditionally on the draws, $B^{-1/2} \sum_b e_b Z_{b,\cdot}$ is a centred Gaussian vector whose covariance is exactly the empirical covariance of the standardised draws, so the distribution of $T^{(\mathrm{mult})}$ approximates that of the max-t statistic without any further refitting [5]. The critical value $q_{1-\alpha}$ is the $(1-\alpha)$ quantile of 20,000 such replicates, generated in chunks so that the $20{,}000 \times B$ normal matrix is never held at once.

Two properties are worth stating because they are the reason the construction is used here. The precision of q is governed by the number of multiplier replicates and not by B, so a design whose fitting budget is fixed by the cost of refitting an entire pipeline can still resolve its critical value; and because the multiplier draws are conditional on the observed Z, the construction inherits whatever dependence the moving-block resampling induced across cells, which is what a family-wise statement over a correlated surface requires. The approximation is asymptotic in the same sense the max-t band is: it needs the high-dimensional central limit theorem to hold over a family whose size is comparable to the draw count, which Section 11 states as an assumption rather than a guarantee.

The Monte Carlo interval for q itself is retained. Writing $[q^-, q^+]$ for the order-statistic interval of the multiplier quantile, a cell is called *resolved* only if its band excludes zero at q^+, so residual Monte Carlo noise leaves cells unresolved rather than resolving them in the wrong direction.

Fieller's construction for the reduction

For $\vartheta = V_1/V_0$ with estimates $(\hat{v}_0, \hat{v}_1)$ and bootstrap covariance $(\hat{s}_{00}, \hat{s}_{11}, \hat{s}_{01})$, Fieller's theorem gives the confidence set

$$\Big\{ \vartheta : (\hat{v}_1 - \vartheta \hat{v}_0)^2 \le z^2 \Big(\hat{s}_{11} - 2\vartheta \hat{s}_{01} + \vartheta^2 \hat{s}_{00} \Big) \Big\},$$

whose boundary solves a quadratic in ϑ with leading coefficient $\hat{v}_0^2 - z^2\hat{s}_{00}$. When that coefficient is positive the set is a bounded interval. When it is zero or negative — which is exactly the case in which the denominator $\hat{v}_0$ is not resolvably different from zero at level z — the set is a half-line, the whole line, or the complement of an interval, and reporting a bounded interval for it is reporting a set that is not the confidence set. `results/s02_fieller.csv` records the kind of every set this paper computes, in its `fieller_kind` column.

The three regret designs

Section 4.7 defines the misreport rate $M(s)$ and the regret $G(s)$ and explains why an average of G taken across instruments has no units. The three designs it names are constructed as follows.

A. Metric-specific. For each scalar instrument m separately, the admissible set is restricted to the cells whose metric axis is at m, and $M(s)$ and $G(s)$ are computed over that restriction under the declared measure. Every reported value carries the name of its unit, and no average is taken across instruments. The only cross-instrument statement the design permits is a count: in how many of the five instruments the *direction* of a conclusion is the same, and which ones it is not — and Section 6.6 reports both, because on this corpus two of those counts are four and not five. The design assumes nothing beyond the measure, and it is what the headline uses.

B. Common operational utility. Net benefit at threshold t,

$$\mathrm{NB}(t) \;=\; \frac{\mathrm{TP}(t)}{n} \;-\; \frac{\mathrm{FP}(t)}{n} \cdot \frac{t}{1-t},$$

is measured in true positives per case at an exchange rate the threshold declares, so values at different thresholds are cardinally comparable and may be averaged. The increment is the difference in net benefit between the two arms at the same threshold, computed only on calibrated probabilities (Section 8) because a threshold read as a cost ratio requires one. The resulting quantity is called *decision* regret rather than predictive-performance regret. The design assumes a distribution over thresholds, which is declared and varied rather than fixed.

C. Partial identification. If a reader insists on one number across instruments, the answer depends on a map ϕ from an instrument's units to theirs. Five are declared, in `s23_regret.UTILITY_MAPS`: the *identity*; division by the instrument's remaining *headroom*, the distance left to a perfect score; division by the increment's *standard deviation* across the surface; a signed *square root*; and a logistic squashing of the standardised increment. Each is increasing in the increment and each divides only by positive quantities, so all five fix zero and the *sign* of an increment is invariant to every one of them; what moves is the relative magnitude an instrument contributes to a mixed average. The conclusion is reported as its *range* over the family together with the part invariant to all of it, which is exactly the part the design declines to identify.

Designs A and C are computed on the full declared surface, since neither needs a bootstrap draw. Design B is computed on the calibrated decision-curve cells of Section 8.

Estimating the decomposition

Write $\mathbb{P} = \bigotimes_i w_i$ for the product measure over the axes. Every subset u of axes has a component g_u given by the Möbius sum over the conditional means,

$$g_u(x_u) \;=\; \sum_{v \subseteq u} (-1)^{|u|-|v|}\, \mathbb{E}_{\mathbb{P}}[V \mid x_v]\,, \qquad D_u \;=\; \mathbb{E}_{\mathbb{P}}\Big[g_u^2\Big]\,,$$

and the components are mutually orthogonal, so $\sum_{u \neq \emptyset} D_u = \operatorname{Var}(V)$ exactly. The first-order and total indices of Section 4.5 are $S_i = S_{\{i\}}$ and $S_{T_i} = \sum_{u \ni i} S_u$ for $S_u = D_u / \operatorname{Var}(V)$.

The components are computed exactly rather than by a sampling estimator, which the complete factorial makes possible. For that product measure the conditional means $\mathbb{E}_{\mathbb{P}}[V \mid x_v]$ are weighted group means over the cell table, one for each of the 2^k subsets v of the k axes; the Möbius sum then gives every g_u and every $D_u = \mathbb{E}_{\mathbb{P}}[g_u^2]$. The implementation forms the group index once with a mixed-radix encoding of the axis levels and reuses it for all 2^k subsets, so the cost is 2^k passes over the cell table rather than 2^k table joins.

Three checks run on every decomposition and are recorded rather than assumed. The disjoint components must sum to the variance, which is asserted numerically on all 304 decompositions. Duplicating a level with its weight split must leave every component unchanged, and does, to 5.6×10^{-17}. Refining the operating-point grid under the same underlying continuous measure must leave the threshold axis's index unchanged, and moves it by 0.004.

Where the inference surface supplies draws, the whole decomposition is recomputed inside every draw, which gives an interval for every index rather than a point; corpus-level medians are bootstrapped with the log–target pair as the resampling unit, because pairs and not cells are the independent objects. No axis is called dominant anywhere in this paper where those intervals overlap.

S3.1. Specification regret: the three designs

A reader of a one-number claim decides whether to adopt the field, and the rule the number supports is *adopt iff* $V_s > 0$. If the specification the reader will actually run is drawn from the same admissible set under the declared measure, two quantities follow. The *misreport rate* $M(s) = \Pr_{s'}[\operatorname{sign} V_{s'} \neq \operatorname{sign} V_s]$ is the probability that the reader's own cell disagrees in sign with the reported one. The *regret*

$$G(s) \;=\; \mathbb{E}_{s'}\Big[\max(0, -V_{s'})\Big]\,\mathbf{1}\{V_s > 0\} \;+\; \mathbb{E}_{s'}\Big[\max(0, V_{s'})\Big]\,\mathbf{1}\{V_s \leq 0\}$$

is the performance destroyed by adopting when the reader's cell is harmful plus the performance forgone by declining when it is beneficial.

$G(s)$ carries the units of the instrument its increments were measured in, so an average across five scalar instruments has no units at all. Three designs are therefore reported, each constructed in Supplement S3: **A, metric-specific**, inside one instrument at a time, which is what the headline uses; **B, common operational utility**, on calibrated net benefit (Section 8.4); and **C, partial identification**, over five declared monotone utility maps fixing zero. The misreport rate has a second difficulty, which Section 6.4 answers rather than notes: a cell whose sign the data do not resolve contributes a coin flip to it. The rate is therefore reported over all cells, over the cells the calibrated band resolves, and weighted by $|V|$, and the article's headline is the second.

Saying that a surface report asserts a sign only where the region is uniform, and therefore misreports nothing, is a definitional win and not a measured one, so the paper compares explicit *decision rules* under the loss above.

Lemma S1 (The decision-optimal collapse)**.** *Let the reader's specification be drawn from the admissible set $\mathcal{S}$ under a weighting w, and let a rule choose one action for all readers. Writing $A = \mathbb{E}_w[\max(0, -V_s)]$ for the harm mass and $B = \mathbb{E}_w[\max(0, V_s)]$ for the benefit mass, expected loss is A under* ADOPT *and B under* DECLINE, *and $B - A = \mathbb{E}_w[V_s]$. Hence expected loss is minimised by adopting exactly when $\mathbb{E}_w[V_s] > 0$: the decision-optimal summary of a specification surface is the* weighted mean increment, *not the sign at any one cell and not the count of positive cells.*

Its assumptions are strong — one action for every reader, a loss linear in a common utility, and a known reader distribution w — and what is scientifically difficult is specifying the utility and the weighting, not proving the identity. The paper therefore reports the identity as a check — the mean rule's in-sample excess regret is exactly zero across every pair and instrument, as it must be — and measures the rules *out of sample* instead (Section 6.6). Two things follow: the conservative rule is wrong, not merely cautious, whenever the harm mass is smaller than the benefit mass it forgoes; and a one-number report is optimal exactly when the sign of its cell agrees with the sign of the mean, which is a coincidence rather than a property of the cell.

S3.2. The pipeline, and what the estimator holds fixed

The pipeline is frozen and identical on every log. Attributes are assigned to roles by the pre-registered rules of Section 5; cases are ordered by their first timestamp; and the split is temporal. The *pipeline* axis spans encodings as well as model families: one-hot logistic regression, frequency-encoded logistic regression, cross-fitted target-encoded [22] histogram gradient boosting, and the same boosting wrapped in isotonic calibration. All four run on the case study's log; the other pairs carry two or three by a rule declared on row count. Because those levels move two things at once, Section 6.2 crosses the family with the encoding where the budget allows it, and the axis is called a pipeline rather than a learner throughout. Registers here run to 3,019 levels, above the cardinality cap of the boosting implementation's native categorical support, which is why the tree learner is target-encoded, cross-fitted inside the training half so that no training row sees its own outcome in its own encoding.

The usual interval for such an increment resamples test rows with the fitted models held fixed. It excludes training-sample variability, model-selection variability, encoder variability, split uncertainty and the temporal dependence between cases, and a design sweep is not a substitute for it: specification sensitivity and sampling uncertainty answer different questions.

Every interval in this paper is instead a *nested moving-block bootstrap*. One draw resamples the training half by moving blocks [17] of length $\lceil n^{1/3} \rceil$, refits every arm of every rung under every learner and every quality mechanism on that resample, and evaluates on a moving-block resample of the test half. Encoders, frequency tables, target encodings, register frequency rankings and calibrators are rebuilt inside the draw. On the simulation of Section 10, where the truth is known by enumeration, the fixed-model interval covers its own estimand a median 92.1% of the time across the worlds and the nested basic interval 93.0%. Every log is also reported under rolling-origin validation — five expanding-window folds — beside the single temporal holdout, and the split is an axis of the decomposition rather than a robustness paragraph.

S3.3. The four interval objects, and what each ranges over

The family must be the family. Within-cell bands, each covering five members, do not cover the several hundred cells a surface-level label ranges over. Four interval objects are therefore declared, and every table says which it prints: POINTWISE, the per-cell interval, with no multiplicity control; WITHIN-INSTRUMENT, max-t over the scalar instruments at one cell, reported only so that the cost of the wider family is visible; WHOLE-SURFACE, max-t over *every* admissible scalar cell of the pair, which is the only family a surface-level claim may use; and DECISION-CURVE, max-t over every admissible net-benefit cell of the pair, declared separately because net benefit and a scalar instrument are not the same scale. The last two are the pre-registered families. The whole-surface family has a median 180 members against the five of the within-instrument family, and its median critical value is 3.33 against 2.35 and the pointwise 1.96.

Family control is **per log–target pair**: every cross-pair statement in this paper is descriptive, and none is a simultaneous inferential claim over the corpus. A replicate is dropped from a family when a resample leaves one of its cells without outcome variation, because a maximum over a moving set is not a maximum; `results/s21_incomplete.csv` names the 2 families where that happens.

S3.4. The planned contrasts, and where a p-value and an interval disagree

A one-sided p*-value and a two-sided basic interval disagree on* PC3*, and the interval decides.* PC3's one-sided p is 0.00100 (Holm-adjusted 0.00250) and its two-sided basic interval, $[-0.00059, +0.01210]$, contains zero. Both come from the same 2,000 draws. The p-value says where those draws lie — their central ninety-five per cent runs $+0.00327$ to $+0.01596$, entirely above zero — while the interval says where the *estimand* lies, obtained by pivoting that distribution about the estimate; the draws sit 0.00193 AUC *above* the estimate, the upward shift Section 4.1 diagnoses, so pivoting moves the interval down by twice that shift and its lower limit crosses zero even though almost no draw does. **A contrast is called resolved only when the interval says so**, and PC3 is not resolved on any of the four cohort-and-target combinations Section 7.1 reports. Every other comparison here is *exploratory*, labelled as such where it appears, and carries no multiplicity correction, because none would be meaningful over a surface enumerated in full.

S3.5. The two estimates of the critical value, and how they differ

And the two estimates of the same critical value disagree, in the direction that matters. The precision argument above is an argument about variance, and the two estimators differ by more than variance. On 15 of the 19 whole-surface families, and on *every* one of the 19 decision-curve families, the multiplier value lies below the entire order-statistic interval of the empirical quantile: the median ratio q_{emp}/q is 1.29 on the surface families and 2.09 on the decision-curve ones, whose medians are 3.87 against 8.10. The direction is anti-conservative — a band built on the smaller value resolves cells the observed maxima do not — and the mechanism is visible in the draws. A Gaussian multiplier reproduces the empirical *covariance* of the standardised draws and therefore only the excursions a Gaussian process makes. These draws are heavy-tailed — Section 10.4 measures how, and finds that the word covers two different mechanisms on the two families — and the asymptotics the construction rests on are asked to hold at 40 to 150 draws over families of several hundred cells.

We therefore report what each estimator resolves rather than asserting one of them. Over the whole-surface families the multiplier band resolves 1,150 cells and the empirical band 995; over the decision-curve families, 3,066 against 613, which is the larger exposure and belongs to the one family the coverage calibration of Section 6.3 does not touch. Section 6.3 carries the surface counts under both and Section 8.3 the decision-curve ones.

S3.6. The measure the decomposition is taken under

The measure the decomposition is taken under. A variance decomposition is taken with respect to a distribution over specifications, and *uniform over the cells I happened to enumerate* is a choice rather than a neutral default: it moves if an analyst adds a pipeline, writes five folds as five levels, or refines a threshold grid. Axis weights are therefore separated from within-axis level weights and every quantity is reported under declared measures (Table S26): EQUAL-LEVEL, every level of every axis equally likely; REFERENCE, half the mass on the reference level of each axis; CONCENTRATED, four fifths on it; and an ENVELOPE over Dirichlet draws of the level weights. Two invariance properties are executed rather than asserted: duplicating a level with its weight split leaves every component unchanged, to 5.6×10^{-17}, and refining the operating-point grid under the same underlying continuous measure moves the threshold index by 0.004.

Indices are estimated quantities: the whole decomposition is recomputed inside every draw the inference surface supplies, corpus-level medians are bootstrapped with the pair as the resampling unit, and no axis is called dominant where the intervals overlap. They are computed within each instrument, in its own units, which is the primary analysis, and on the *headroom* scale $V_s/(1 - m(B))$, which is comparable across instruments and lets the metric enter as an axis. The second is a sensitivity analysis rather than a uniquely meaningful decomposition, because of the following.

Proposition S1 (The decomposition inherits the metric's scale)**.** *Let ϕ be strictly increasing with $\phi(0) = 0$. The first-order sensitivity indices of V and of $\phi(V)$ over the same design need not agree, and there exist designs and ϕ under which one axis has the larger index on one scale and a different axis on the other.*

The 3×3 construction is in Supplement S2. This is the same phenomenon as Remark 1 one level up: just as the reduction is not identified without the metric, "which choice matters most" is not identified without the scale on which the increment is expressed, so the manuscript never says an axis dominates without naming it.

S3.7. Why refitting inside the draw breaks the percentile interval

Refitting inside the draw breaks the percentile interval. A bootstrap resample contains about $1 - e^{-1}$ of the distinct levels of a high-cardinality categorical, so with 3,019 register levels every refit inside every draw sees a sparser register than the real training half and returns a smaller increment. The bootstrap distribution is then displaced relative to the estimate, and a percentile interval taken from a displaced distribution is a correct interval for the wrong quantity: it misses no matter how wide it is. On this corpus the displacement is small — negative in 56% of cells, by a median -0.0001 AUC — but it bites in the regime the mechanism predicts: on the sparse world of Section 10 the percentile interval covers 37.2% against a nominal 95%, *worse* than the fixed-model interval refitting was meant to improve on. Every interval here is therefore the *basic* (pivotal) interval $[\,2\hat{V} - q_{1-\alpha/2},\ 2\hat{V} - q_{\alpha/2}\,]$ [10], which pivots the displacement out; simultaneous bands are recentred by the same shift, which is a property of the draws and so corrects a whole family at once (Table S37, and Supplement S6 for the diagnosis).

S4. Which layer of the register pays

A register has layers. The affected configuration item is the finest; its type, its subtype and the service component it belongs to are deterministic coarsenings, and a buyer choosing how much of a register to populate is choosing between them. The registered rules give coarsenings their own role (S1, amendment 4) so that this question can be asked without the coarsenings contaminating the intake block.

Which cell the ladder is on, and why it does not reconcile with the decision-time table. Section 7.4 reports the item as worth +0.257 AUC over the intake block and Table 8 reports +0.1838 over the same block, and a reader who tries to reconcile the two cannot, because they are different cells. This ladder is computed on the *registered* cohort of Section 5 — 46,606 cases, 13,982 in test — under the registered handover rule, one-hot logistic regression at the fixed temporal holdout, against the intake block at base AUC 0.641 on the reassignment question and 0.618 on duration. Table 8 is on the estate's own cohort and its reassignment target. Section 7.1 is the reason the two differ: they are the two analyses whose answers about this register disagree, and neither number is the other one mismeasured. What the ladder compares is layers *against each other* at one cell, which is the comparison a programme is choosing among, and that comparison is unaffected by which of the two cohorts it is made on.

Table S1 reports the increment of each layer over the same baseline, and of the item over each layer. **Which layer pays is a property of the target**

log	target	level	levels from coarse	cardinality	n	n test	base auc	auc	gain over base
Helpdesk	duration	service_type	0	3	4580	1374	0.629	0.629	0.001
Helpdesk	duration	support_section	1	7	4580	1374	0.629	0.629	0.000
Helpdesk	duration	product	2	21	4580	1374	0.629	0.638	0.009
Helpdesk	duration	product marginal over support_section	-1	21	4580	1374	0.629	0.638	0.009
BPIC19	duration	case:Spend area text	0	20	248445	74534	0.619	0.620	0.000
BPIC19	duration	case:Vendor	1	1944	248445	74534	0.619	0.672	0.052
BPIC19	duration	case:Item Category	2	4	248445	74534	0.619	0.620	0.000
BPIC19	duration	case:Item	3	490	248445	74534	0.619	0.621	0.002
BPIC19	duration	case:Item marginal over case:Item Category	-1	490	248445	74534	0.620	0.621	0.002
BPIC14	handover	CI Type (aff)	0	13	46606	13982	0.641	0.775	0.134
BPIC14	handover	CI Subtype (aff)	1	65	46606	13982	0.641	0.804	0.163
BPIC14	handover	CI Name (aff)	2	3019	46606	13982	0.641	0.898	0.257
BPIC14	handover	CI Name (aff) marginal over CI Subtype (aff)	-1	3019	46606	13982	0.804	0.913	0.109
BPIC14	duration	CI Type (aff)	0	13	46606	13982	0.618	0.662	0.045
BPIC14	duration	CI Subtype (aff)	1	65	46606	13982	0.618	0.662	0.045
BPIC14	duration	CI Name (aff)	2	3019	46606	13982	0.618	0.747	0.129
BPIC14	duration	CI Name (aff) marginal over CI Subtype (aff)	-1	3019	46606	13982	0.662	0.749	0.087

Table S1: The increment of each layer of the register over the same baseline, on the log–target pairs the registered rules of Section 5 admit. 9 rows on excluded pairs are not printed. *Columns.* levels from coarse is the layer's depth, counting from the coarsest layer at 0 inward; the value -1 is not a depth but marks the *marginal* row, which adds the finest layer on top of the second finest. base auc is the AUC of the model this row is measured against — the intake block B_0 with every register layer removed on a layer row, and B_0 together with the second-finest layer on the marginal row. auc is the AUC of that same model with this row's layer added, and gain over base is the difference of the two. Because the marginal row's baseline is not B_0, its gain is not comparable with the rows above it and is not meant to be: it answers what the finest layer is worth once the coarser one is already in the model.

and of the log, and not of the register. On the case study's estate the coarse layers are *not* close to free: the item's type alone reaches 52% of what the item is worth over the intake block on the reassignment question and 35% on the duration question (Section 7.4), and what survives as a claim about the finest layer is its *marginal* contribution over the coarse ones — +0.109 and +0.087 AUC. On the two other logs in the table the opposite holds: every coarse layer is worth essentially nothing — at most +0.001 on Helpdesk and −0.002 on BPIC19 — while the finest is worth +0.009 and +0.050. **So a programme cannot read a layer policy off another estate's ladder**, which is the same conclusion this paper reaches about the increment itself, one level down. None of it transfers to any other task and none is claimed to.

S5. Rolling-origin results, per fold

The single temporal holdout is one split. Table S2 reports the increment at the reference cell under each of the five expanding-origin folds, per log–target pair, so that a reader can see how much of the holdout's answer is the

log	target	rolling1	rolling2	rolling3	rolling4	rolling5
BPIC13_closed	handover	-0.019	-0.011	-0.024	-0.012	0.015
BPIC13_incidents	duration	0.001	0.027	0.051	0.030	0.007
BPIC13_incidents	handover	0.030	0.038	0.049	0.032	0.032
BPIC14	duration	0.076	0.100	0.083	0.109	0.112
BPIC14	handover	0.092	0.121	0.152	0.155	0.141
BPIC15_1	duration	0.034	0.030	-0.020	-0.018	0.008
BPIC15_1	handover	0.001	0.010	0.001	0.005	0.004
BPIC15_3	duration	0.166	0.033	0.050	0.031	0.026
BPIC15_3	handover	0.031	0.020	0.037	-0.023	0.007
BPIC15_4	duration	-0.040	-0.004	-0.004	0.031	0.023
BPIC15_4	handover	0.000	-0.018	0.000	-0.032	0.038
BPIC15_5	duration	0.050	0.052	0.024	0.026	0.052
BPIC17	duration	0.003	0.002	0.005	0.010	-0.007
BPIC19	duration	0.044	0.053	0.059	0.018	-0.004
Helpdesk	duration	0.003	-0.004	0.016	-0.039	0.000
RoadFines	duration	0.024	0.010	0.021	0.033	0.041
Sepsis	duration	0.130	0.159	0.154	0.129	0.186
UCI498	duration	-0.007	-0.011	-0.008	-0.006	-0.010
UCI498	handover	0.003	-0.003	0.001	-0.003	-0.005

Table S2: The increment at the reference cell under each rolling-origin fold.

holdout's.

The split axis carries a median 18.9% of the variance in the increment (Table S8) — the largest of the four first-order indices on this corpus, which is worth stating because the split is the axis an evaluation paper is most likely to fix without comment. Reporting a single holdout is therefore not enough, and where a fold disagrees with the holdout in sign the pair's region label in Table 4 reflects it.

S6. The noisy world, and two explanations that were wrong

The simulation of Section 10 reports that every interval construction fails on the noisy world: coverage 7.0% against a nominal 95%, with the estimator sitting -0.0353 below V_{limit} and a mean interval width of 0.0367. A gap the size of the whole interval is not something a resampling scheme repairs, and the natural write-up was that the estimator is biased in that world and the interval inherits the bias.

That write-up would have been wrong, and the two experiments below are what stopped it. Both were designed so that either could refute the bias reading, and both did.

The gap against sample size. A finite-sample bias closes as n grows. The increment was estimated at six sample sizes from 2,000 to 64,000 in every world, at 30 replicates each, with the penalty held fixed. The sparse world behaves as a finite-sample gap must: +0.0118 at $n = 4{,}000$ falling to +0.0006 at 64,000. The noisy world does not: +0.0361 at $n = 4{,}000$ and still +0.0329 at 64,000. The sparse world is the control that makes the noisy result mean something — without it, "the gap did not close" could be a statement about the experiment rather than about the world.

The gap against the penalty. A fixed L_2 penalty is not scale-free, so it can masquerade as bias: it shrinks coefficients materially at $n = 4{,}000$ and barely at $n = 120{,}000$, which would make a large-sample fit a different specification rather than the same one at its limit. Holding n at 4,000 and weakening the penalty by five orders of magnitude moves the noisy world's gap from +0.0361 to +0.0362 — that is, not at all.

What was actually wrong. Neither explanation surviving is what sent us to examine the target rather than the estimator, and the target was wrong. The simulation enumerated the generator's cells, carrying the *true* register value, while in the noisy world the estimator only ever sees a value that is replaced by a uniform draw with probability 20%. Both estimands were therefore computed on a population the estimator does not sample from. Marginalising the noise exactly (Section 10) moves V_{limit} from +0.1135 to +0.0804. The estimator's mean was +0.0781 throughout. There was no bias; there was a wrong estimand in our own simulation, and it took two arms of evidence against everything else to find it.

The same two experiments, after the correction. Both arms were re-run against the corrected target, and there is nothing left for them to explain: the noisy world's gap is +0.0025 at $n = 4{,}000$ and −0.0007 at 64,000. The sparse world's gap is unchanged in both runs, because the correction touches only the noisy world, which is what makes it a control rather than a coincidence. Coverage on the noisy world is 94.8%. `scripts/s18_bias_scaling.py --legacy-target` regenerates the pre-correction figures above, so the

world	fixed-model, basic	fixed-model, percentile	nested, basic	nested, bias-corrected	nested, percentile	MC half-width
drift	0.935	0.915	0.910	0.915	0.975	0.030
imbalanced	0.910	0.910	0.935	0.950	0.940	0.030
linear	0.920	0.900	0.915	0.915	0.945	0.030
noisy	0.930	0.930	0.920	0.925	0.920	0.030
nonlinear	0.935	0.925	0.935	0.940	0.950	0.030
sparse	0.660	0.635	0.925	0.675	0.420	0.030

Table S3: Simulation: coverage of the estimator's own limit at the nominal 95%, by world, for the fixed-model and the nested bootstrap under the percentile, basic and bias-corrected constructions. The nested percentile interval is the one that fails, on the sparse world, and the basic construction is the repair; the bias-corrected percentile is not. The last column is the Monte Carlo half-width of every coverage in the row at this replicate count, so a reader can see which differences the simulation resolves. The oracle rows are in results/s10_coverage.csv.

evidence for this appendix is reproducible rather than quoted from a log we happen to have kept.

The general lesson is the one this paper makes elsewhere about the design space: a quantity is defined relative to a declared population, and a simulation's population is a declaration like any other. A quality mechanism that acts on the data — which is exactly what the register-quality axis of Section 3 does on real logs — changes what the estimand is, not merely how well it is estimated. The register-quality results of Section 7.3 are reported against the degraded register for that reason, and the correction register in the archive records this as the round's largest error of its class.

The two estimands, and the marginalisation

Two estimands are separated, because under misspecification they are not the same. V_{oracle} is the increment between the two Bayes-optimal scoring rules — what the register is worth in the world. V_{limit} is the increment between the two fitted pipelines' limits — what the estimator estimates. A resampling interval can only cover the second, and reporting its coverage of the first as though it were the estimator's coverage is precisely the error the earlier simulation made. Both are reported in Table S3, which is the 200-replicate run this diagnosis was made on. Table S37 is the 1000-replicate run every coverage number in the main text comes from; the two agree on every qualitative statement in this appendix, and the diagnosis is reported against the run that produced it rather than restated against a later one.

The truth has to be the truth the estimator sees. Both estimands are computed by enumerating cells, and in the noisy world the cells the generator writes down are not the cells the estimator observes: a register value is replaced by a uniform draw with probability 20%, so the observed cell (b, g, j) is a mixture over the true ones, with mass and outcome rate

$$\begin{aligned} w(b,g,j) &= \sum_f \Pr(b,g,f)\ \Pr(j \mid f), \\ \Pr(y{=}1 \mid b,g,j) &= \frac{1}{w(b,g,j)} \sum_f \Pr(b,g,f)\ \Pr(j \mid f)\, p(b,g,f), \end{aligned}$$

where $\Pr(j \mid f) = (1-r)\mathbf{1}\{j = f\} + r/K$. Enumerating the *true* cells instead scores the estimator against a population it never samples from, and the first version of this simulation did exactly that: it reported a coverage of 7.0% on the noisy world and a bias of -0.0353 that did not exist. S6 gives the two experiments that refuted every finite-sample explanation for that apparent bias and thereby sent us to look at the target. The marginalisation is checked against a Monte Carlo sample, on standardised residuals, at the start of every simulation run.

S6.1. The six simulated worlds and the two estimands

A simulation that fits a logistic data-generating process with logistic regression tests an unusually favourable arrangement. This one uses six worlds — four of them misspecified for the estimator — at 1,000 replicates each. The covariate space is finite, so the joint distribution of cell and outcome is enumerated and the population AUC of any scoring rule is computed in closed form rather than simulated. The worlds are a logistic surface, a nonlinear one with an interaction and a threshold the logistic model cannot see, a drifting one, a sparse register with 200 levels, an imbalanced outcome at prevalence 0.02, and a noisy register in which a share of values are wrong.

Two estimands are separated because under misspecification they differ. V_{oracle} is the increment between the two Bayes-optimal scoring rules — what the register is worth in the world. V_{limit} is the increment between the two fitted pipelines' limits — what the estimator estimates. A resampling interval can only cover the second, and reporting its coverage of the first as though it were the estimator's coverage is a category error. Both are reported in Table S37. Where a register value is corrupted the cells the generator writes down are not the cells the estimator observes, so the truth has to be marginalised over the corruption; the two experiments that found that and the check that guards it are in Supplement S6.

S6.2. Where the refit helps, and where the percentile interval fails

Where the refit helps. The nested basic interval covers at least as well as the fixed-model percentile interval on 6 of the 6 worlds, and its median across them is 93.0% against the fixed-model interval's 92.1%. Both fall a little short of nominal on most worlds, by more than the Monte Carlo resolution.

Where the percentile interval fails, and why. On the sparse world it collapses to 37.2%, and the *nested* percentile interval is worse there than the fixed-model one. The cause is Section 4.1: a bootstrap training resample holds about 63% of the distinct register levels, so every refit sees a sparser register and the bootstrap distribution shifts below the estimate. The basic interval pivots that shift out and restores coverage — 89.9% — and costs nothing where there was nothing to repair, at a median 93.0% and a minimum 89.9% across all six worlds. The bias-corrected percentile does *not* repair the sparse world (63.1%), which is why every interval and band in this paper uses the basic construction. The m-out-of-n subsampling interval, the standard alternative in exactly this regime, does not win either: 72.8% on the sparse world and a median 75.8% across the 3 worlds it was priced on.

S7. The worked example on UCI Adult

The standard is not about event logs and not about configuration databases, and the quickest way to show that is to run it on something else. UCI Adult (Census Income) predicts whether a person's income exceeds a threshold from 30,162 records. Suppose an analyst wants to report what `occupation` is worth. It is the expensive field: a survey has to ask about it and code it, where age, sex, race and country come with the sampling frame and education comes with an enrolment record.

Worth compared with what? Against the frame alone, against the frame plus education, or against the frame plus education plus hours worked. Measured how, at which operating point, and with the register how complete. Over the resulting 480 cells the reduction runs 0.200 to 0.947. The baseline axis and the metric axis carry the same first-order share of the variance in the increment to three figures — 25.9% and 25.9% — against 7.2% for the register population, and a one-number report misstates the sign for a mean 24.9% of the other cells. That the two lead axes tie is a property of this dataset and not a general one; it is the sort of thing a surface reports and a single number cannot.

`examples/worked_example.py` produces all of it in 121 statements against the package's public interface, and ends by calling `float()` on the surface so a reader can see what the refusal looks like.

S8. The verification harness

Every number in the manuscript is a macro, defined in the generated file `paper/numbers.tex` that `scripts/make_numbers.py` writes from `results/*.csv`. The manuscript contains no numeric literal outside that file and the generated tables. A number therefore cannot disagree between the abstract, a table and the conclusion: there is one of it.

That removes one class of defect and creates another — a wrong generator makes the whole manuscript wrong *consistently*, and nothing looks odd — so the architecture comes with seven controls.

The generator. `make_numbers.py` emits a visible **??** for any macro whose source file is absent, and `--strict` exits non-zero if one remains. A partial build is therefore visibly partial and cannot silently drop a claim.

The verifier. `scripts/verify_numbers.py` re-derives each checked macro from the result file with code written against the *definition* rather than against the generator, and compares. It also enforces 25 consistency conditions that no single macro expresses — relations that must hold between numbers even when each of them is individually correct. Among them: that every register-quality mechanism passed its executed train-only check; that no surface is labelled uniformly beneficial while carrying an unresolved cell, nor conditionally harmful while carrying a beneficial one; that every region label is one of the six the manuscript defines and that the labels partition the pairs; that Remark 1's sweep still holds the AUC reduction at zero; that the disjoint functional-ANOVA components sum to one on every decomposition; that no planned contrast reports a p-value the draw count cannot produce, and that its Holm adjustment reproduces from the raw values; that no rank-based instrument's reduction moves under a monotone recalibration, and that every certificate for a non-rank-based one exhibits two ratios that actually differ; that the cells a region label ranges over are exactly the cells the simultaneous family that licenses it controls, and that every decision rule is compared over the declared regret population rather than over whatever the file happened to hold; that the case study's split partitions its cohort; and that the adjudicated set of the practice pilot is exactly the screened-in set. The partition condition is the one that catches a region label the analysis emits and the manuscript does not define, which is how it was found.

The corruption suite. `scripts/s13_attack_numbers.py` is the verifier's own regression suite. It corrupts one thing at a time — a value in a result file, a value in `numbers.tex`, a deleted macro, a train-only check flipped, a region label made inconsistent with its counts, an adjudication file made inconsistent with the coding file — and requires the verifier to notice. Every corruption is applied to a copy in a temporary directory; the suite refuses to start if a previous run left its lock behind, because in an earlier round a corruption suite was killed mid-flight and a corrupted manuscript reached a commit.

A corruption of a *result* file is checked by regenerating the macros from the corrupted copy and then verifying, which is the only version of that test that means anything: without the regeneration the suite asks the verifier about a `numbers.tex` built from clean results, so any quantity the verifier re-derives from a different file than the generator reads is untouched and the corruption passes. One corruption did exactly that, silently, for a round.

The condition on the set, not on the number. Every check above compares one macro with one source file, and a whole class of defect passes all of them: two macros, each perfectly consistent with its own source, that the prose calls by the same name. That is not hypothetical. It is how a case study's increment came to be printed with two values and two signs, and no check in this repository could have seen it.

The repair is a declaration rather than a computation. The verifier `round21_verify.py` carries an ESTIMANDS table: one entry per quantity a reader would name in one phrase, listing the macros that may hold it. Two members of an entry that disagree is a failure; a quantity that legitimately takes two values under two specifications gets two entries with different names, and the manuscript must then use those names. Corruption `G1` in the suite is the test: it moves a value in one result file and lets both the generator and the verifier read the moved value, so every macro-against-source check passes and only the condition on the set fails. **Write the check on the name, not on the number.**

The provenance record. `results/provenance.json` holds the SHA-256 of every analysis script at the moment its outputs were last accepted, with a note saying why, and `make_numbers.py --strict` refuses to certify a manuscript whose numbers come from a script that has changed since. This exists because `--strict` used to pass on a build reading a two-replicate smoke test that had been run under the wrong seed and against an estimand corrected an hour later: every macro resolved cleanly, to a wrong number. A missing file is loud, and a stale one is silent. Modification time cannot answer the question — changing a script's output format does not make its numbers wrong — so acceptance is a deliberate act with a recorded reason.

The compliance linter. `scripts/texlint.py` carries 16 checks: the abstract length, the keyword count, the highlight lengths against their own stated counts, the presence of all five required statements, the absence of any numeric literal in the prose, the absence of a declared set of rhetorical sentence-openers, that every `\input` target exists, that a spelled-out count matches the list it introduces — whether that list is a run of identifiers or a `description` environment — and that no manuscript part, generated table or analysis script carries a stray control character.

Seven of them exist because each caught a defect that had already reached a compiled PDF of *this* manuscript: a spelled-out count that did not match the list it introduced; the word *Appendix* written before a reference that already supplies it, on every appendix reference at once; a macro swallowing the space after it and printing `42.0%--- and`; a macro carrying words quoted inside math mode and printing `18percentagepoints`; a sentence's tail left behind by a scripted edit and printed twice; a *backslash escape interpreted by a scripted edit*, which had written `\ref{x}` into the file as a carriage return followed by `ef{x}` and `\nMacro` as a newline followed by `nMacro`; and a spelled-out count introducing a `description` list of a different length — which is the first defect in the form the first check could not see, and which was in the paragraph above this one, announcing five controls before listing six.

Those print as garbage or as a wrong word, produce no wrong number, and every checker in this repository passed on the build that carried them. Each signature is exact — a carriage return that is not a line ending, a line beginning with the tail of a control word, a numeral word whose list is a different length — so each is now a check rather than a thing to remember. All seven were found by *reading the rendered pages*, which is the check no file in this repository replaces.

The claim registry. `scripts/claim_registry.py` writes one row per macro: its value, every manuscript location that uses it, the generator file and *line* that defined it — captured from the calling frame, so it cannot drift — the result file that line reads, and whether the verifier re-derives it independently. Because the manuscript contains no numeric literals, the registry is exhaustive by construction: any number a reader can see is a macro, and every macro is a row. `--check` fails a build in which a macro the manuscript uses is unresolved.

The registry also answers the question a reader should ask of the paragraph above: *how many* macros are independently re-derived, not merely how many could be. Its summary row counts them, separately for the *headline* macros — the ones the abstract, the introduction and the conclusion print — and the count is deliberately not quoted here, because it is a property of a build and belongs in the file rather than in a sentence. It is not all of them, and the ones it is not are visible in the same file. Writing the missing re-derivations is also how the largest arithmetic error in the register was found: a range taken over four rows of a table, three of which are corpus medians and one of which is a single pair.

What the harness does not do. It guards numbers thoroughly and prose only where a guard was written by hand. The correction register in the archive groups this project's own errors by class, and its largest class is sentences in which every printed number was correct and the relation asserted between them was not. The generated-macro discipline makes one sub-class of that structurally impossible — the same quantity printed differently in two places — and does nothing about the rest. Holes have been found in this project's checkers in every round including this one — a corruption that passed because nothing regenerated between it and the check, a strict mode that certified a manuscript built from a superseded run, a compliance rule that reads digits and not words — and every one of them was found by a corruption suite, by a provenance record, or by reading the output, never by the checker itself.

The clearest limit is Class G of the register: a quantity computed correctly, printed correctly, and consistent everywhere it appears, while the *definition* it refers to is not the one the estimator addresses (S6). Every guard described above passes such a manuscript, and would have passed it silently. What caught it was constructing two experiments that could each have confirmed the comfortable reading and having both refuse to. No checker in this repository would have done that on its own.

S8.1. The inference surface, the draw counts and the pipeline axis

The inference surface is smaller than the declared surface, and the grid is not uniform across the corpus. Every point estimate comes from the full declared space; every *band* comes from the inference surface, a median 12.5% of each pair's computational surface. It is a subset of the declared surface cell for cell and the share of positive cells differs between them by a median 0.067, but a study with compute to spare should equate the two. The same budget shows in two further places. The draw counts run 40 to 150 as declared and 33 to 150 effectively, because a replicate that leaves any cell of a family without outcome variation is dropped from that family; the two families that lose the most sit on logs that also supply much of the sign disagreement of Section 6.4, so the headline's largest contributors are also its noisiest. And the pipeline axis carries four levels on the case study's log and two or three elsewhere, by a rule declared on row count before the runs, so a sensitivity index computed on one pair is not strictly comparable with one computed on another.

S9. The coverage calibration

Section 6.3 reports every region label and every robustness index under a critical value calibrated at the pair's own ratio of register cardinality to training row count. This section gives the construction the article summarises.

S9.1. The inflation factor has a closed form

Write the basic interval of Section 4.1 as $[\hat{V} - L,\ \hat{V} + U]$. Inflated by a factor c it becomes $[\hat{V} - cL,\ \hat{V} + cU]$, and it covers the truth ϑ exactly when

$$c \ \geq\ \max\left\{\frac{\hat{V} - \vartheta}{L},\ \frac{\vartheta - \hat{V}}{U}\right\}.$$

The smallest c attaining nominal coverage at level $1-\alpha$ over a set of replicates is therefore the $(1-\alpha)$ quantile of that ratio across them, and its Monte Carlo interval comes from the same order statistics. No search over candidate factors is needed, and the estimate inherits the replicate count directly: the plane carries 29 cells at 500 replicates each, over K/n from 0.0011 to 0.500, on two data-generating worlds rather than one (Table S41).

S9.2. A monotone fit, not a parametric one

The dependence of c on K/n is well summarised by a log-linear fit whose slope is 0.102, with a standard error of 0.005: the widening a pair needs grows with its register's cardinality relative to its training set, as the mechanism of Section 4.1 predicts. That fit is quoted because it is the interpretable summary, but it is *not* what the corpus is calibrated with. The parametric fit sits below the measured factor at the smallest ratios, and using it would under-correct exactly the pairs whose bands are narrowest — the pairs where an anti-conservative band does the most damage. The corpus is therefore calibrated with an isotonic regression through the measured factors, which is monotone by construction and interpolates the measurements rather than smoothing through them. Table S41 prints the two curves side by side.

Three candidate curves, on the criterion that chose the monotone one. The widened plane makes $\log n$ a significant term (Section 6.3), so the obvious repair is a two-covariate parametric fit in $\log(K/n)$ and $\log n$. It is not one. On the plane's fitted cells the weighted residual sum of squares is 253 for the ratio-only fit, 185 with $\log n$ added, and 52 for the monotone curve; and on the criterion that matters — how many cells the curve sits below their own measured factor's lower confidence limit, which is the anti-conservative direction — the two parametric fits are alike at 8 and 8 cells against the monotone curve's 2. Adding n to a parametric curve improves the fit and does not improve the object. The monotone curve is retained and its residual in n is reported as a limitation rather than fitted away.

S9.3. What the calibration does and does not correct

The factor is estimated on a *pointwise* interval in a simulation and applied to a *simultaneous* band on real logs. It corrects the scale of the studentised statistic, which is the channel the plane identifies. It does not correct a change in the shape of the maximum's distribution, and it assumes that the simulated worlds' dependence on K/n transfers to logs whose dependence structure and signal are not the simulation's. Because that transfer is an assumption, Section 6.3 reports the corpus counts at three settings — nominal, calibrated, and every factor pushed to the upper end of its own Monte Carlo interval — and Section 11 carries the transfer as a limitation.

S9.4. The plane's reach in n, and where the factor is undefined

The factor's reach in n, and the two things widening the plane found. The plane is fitted at training sizes 500 to 180,000 and this corpus runs 735 to 176,213, so all 19 of the 19 pairs lie inside the range the factor is identified on. That is the arrangement a calibration needs, and it is not the end of the matter.

K/n is not sufficient, and a plane wide enough in n can now show it. Adding $\log n$ to the fit gives a coefficient of 0.067 at $t = 8.3$: at a fixed ratio the widening a pair needs grows with its sample size. The design that isolates it is the ladder, three ratios held fixed while n moves over two orders of magnitude, and along it the measured factor moves by 0.22 — 2.8 combined Monte Carlo standard errors — while coverage itself moves by 21.7%. The applied curve is a function of the ratio alone, so it carries that dependence as a residual: it sits below the measured factor's own lower confidence limit on 2 of the plane's cells, by at most 0.18, and both are large-n cells. A two-covariate fit in $\log(K/n)$ and $\log n$ does not repair it — it is worse than the monotone curve on every cell-weighted criterion (Supplement S9) — so the residual is reported rather than fitted away.

And on 4 of the 33 cells the factor does not exist at all. The inflation factor is a widening *about the point estimate*, and Section 4.1 shows that past a crossing in n the basic interval stops containing the point estimate. Where it does, there is no widening about $\hat{V}$ that attains nominal coverage and the quantity this calibration estimates is undefined. All 4 such cells are at 50,000 training rows or more. The admission rule — a cell enters the plane when the interval straddles its estimate in at least 0.90 of replicates — decides nothing: the rejected cells reach 0.078 and the admitted ones start at 0.976, so any threshold between those two gives this plane. Section 11 carries what that leaves the corpus's largest pairs.

Under the calibrated band the region counts are: conditionally beneficial, 9; sign-changing, 3; conditionally harmful, 1; unresolved, 6; and the median robustness index is 0.017.

S9.5. The (n, K) grid, cell by cell

The ratio K/n captures most of the dependence, and not all of it. Across the plane the basic interval covers a median 83.3% and a minimum 21.3%, the last at a ratio of 0.377. Below $K/n = 0.050$ its coverage lies between 88.7% and 90.7% — short of nominal everywhere, by about one Monte Carlo standard error at 150 replicates — and it has fallen below three quarters by 0.125. The ratio is the right first-order summary — the mechanism of Section 4.1 is why, and varying n alone would not have exposed the regime the corpus occupies — but it is not sufficient, and the plane says so in two places. The two cells closest to each other in ratio — a register of 200 levels at 2,000 training rows, and one of 1,000 at 8,000 — cover 0.780 and 0.613, a gap of 3.2 combined Monte Carlo standard errors; and holding the register at 1,000 levels while moving the training half from 2,000 rows to 8,000 moves coverage from 0.453 to 0.613, so n carries signal of its own. The calibration of Section 6.3 is a function of the ratio alone and therefore inherits this residual, which Section 11 states.

Where this corpus sits, and what is done about it. The median log–target pair here has $K/n = 0.106$, the range runs 0.001 to 0.324, and 10 of 19 pairs are above a tenth. Most of this corpus therefore sits at or beyond the ratio at which coverage starts to fall, and a nominal band on those pairs is *narrower* than a 95% interval should be — the anti-conservative direction, which inflates the count of resolved cells. The plane is dense enough to calibrate against, and Section 6.3 does.

S9.6. The degenerate decision-curve cells

What is wrong with the decision-curve families is not a tail. At a threshold where both arms treat every case alike the net-benefit difference is zero by arithmetic rather than by estimation, and 841 cells of the corpus's curve families are exactly zero in at least nine draws of ten — up to 17.2% of a single family, against 0 such cells anywhere in the surface families. The admission rule keeps any cell whose spread exceeds a floor in absolute units, so such a cell survives and standardising it by that spread sends its studentised value to the largest value a deviate standardised over B points can take. They are what makes q_{emp} large on those families: excluding them moves the median ratio q_{emp}/q from 2.09 to 1.80. It does *not* dissolve the disagreement — q still lies below the empirical interval on 19 of the 19 curve families and 15 of the 19 surface ones once they are gone — and Table S40 reports the regime with and without them because Section 4.3's count of what q_{emp} resolves is, on those families, partly a count of arithmetic.

S9.7. The region counts at four calibration settings

The counts are reported at four settings, and the ordering is stable across all of them. At $c = 1$ — the nominal band, no calibration at all — the corpus resolves 1,150 cells at a median ρ of 0.117; under the calibrated factors, 896 cells at 0.017; pushing every factor to the upper end of its own Monte Carlo interval, 833 at 0.011; and under the *empirical* critical value of Section 4.3 rather than the multiplier one, 995 against the multiplier's 1,150. That last comparison is not a calibration setting but a different estimator of the same quantity, and it moves the resolution by more than the calibration does. What moves across all four is how much resolution the corpus has; what does not move is that no pair reaches $\rho = 1$. The transfer from a pointwise interval in a simulation to a simultaneous band on real logs, and the choice between the two critical-value estimators, are both assumptions, and Section 11 carries them as limitations.

S10. A glossary, with the case study's own value for every term

Every term the article uses in a technical sense, with what it means and what it is on the case study's log. The article defines each where it introduces it; this is the lookup.

term	what it means	on the case study's log
register f	the recorded field whose worth is in question: high-cardinality, maintained, reused across cases	the affected configuration item
free field g	a resource-like field that is recorded anyway and that the register has to beat	the assignment group
intake block B_0	what is known when the case arrives, excluding the register and the free field	category, impact, urgency, priority
rung	one baseline in the nested ladder $B_0 \subset B_1 \subset \cdots$ the increment is measured against	intake + group + knowledge reference
instrument	the scalar metric an increment is read in; five are declared and they are not on a common scale	ROC AUC
cell	one complete assignment to every varied axis: a point of the surface	one-hot logistic, holdout, clean register, intake + group, AUC
specification surface	the map from cells to increments over a declared design space	the 4,800 cells of Figure 1
declared / computational / inference surface	what is admissible, what was computed, and what carries bootstrap draws — three denominators, audited apart	29,040, the same, and 3,900
resolved	the cell's simultaneous band excludes zero, so the data determine the sign of that cell's increment	197 of 5,808 AUC cells corpus-wide
family $\mathcal{F}$	the set of cells a claim quantifies over, which is what a simultaneous band must cover	a median 180 cells per pair
resolution region	the label a surface earns from its resolved cells: uniformly or conditionally beneficial or harmful, sign-changing, unresolved	sign-changing
robustness index ρ	(beneficial − harmful) / $\lvert\mathcal{F}\rvert$, always printed with the counts behind it	see Table 6
MPID	the minimal practically important difference, declared before any disagreement is counted	0.01 AUC

Table S4: THE VOCABULARY, IN ONE PLACE. Every term this paper uses in a sense a reader could not guess, with the case study's own value for it. The terms are defined again where they are first used; this table exists so that a reader who meets one in Section 6 does not have to find that place.

S11. Which halves a register-quality mechanism is applied to

A degraded register is a state of the world and not a training accident, so **the mechanism is applied to both halves of the data**: the same values are missing, stale, corrupted or split at fit time and at evaluation time, and the increment measures a model built on a degraded register and deployed against one. What is estimated from the *training half alone* is the mechanism's *parameters* — which values fall in the long tail, which are common, the frequency table a corruption draws from, the set of identities a reconciliation failure splits — so that no test outcome and no test-half content can reach the counterfactual.

That property is executed rather than asserted: the test half is perturbed, the mechanism is re-applied, and the training half's degraded values are required to be identical, on every mechanism and level. The check runs on every mechanism and every level in the design space and is one of the conditions `verify_numbers` enforces (Supplement S8).

The distinction matters because the two designs measure different things. A mechanism applied to the training half alone would degrade the model's inputs and leave the evaluation clean, which measures a train–test mismatch: how much a model built on a bad register loses when deployed against a good one. That is not the question a register's owner is asking. The owner's register is bad at both moments, and the counterfactual that answers the question degrades both.

S12. A prefix axis, on one log

The article predicts at one moment — case creation — from the attributes recorded there, and Section 11 states plainly that prefix length, prefix bucketing and sequence encoding are therefore not axes of $\mathcal{S}$ and cannot be. That limitation stands. What it leaves open is the question a reader from predictive process monitoring will ask next: whether the paper's central claim — that the moment of prediction is an *argument* of the estimand rather than a preliminary to it — is a property of that literature's own axis as well, or an artefact of predicting at creation. This section answers that on one log.

The construction, declared before it was run. The log is BPIC14, the case study's own, and the register is the affected configuration item. At prefix k the analyst stands after the case's k-th assignment event. Three things follow from k and are not chosen separately. The **population** is the cases

prefix	n	prevalence	register levels	base auc	with auc	V	interval	resolved
0	46606	0.945	3019	0.641	0.898	0.257	[+0.233, +0.281]	True
1	46605	0.945	3019	0.733	0.900	0.166	[+0.145, +0.203]	True
2	11080	0.722	1567	0.711	0.775	0.064	[+0.047, +0.088]	True
3	6583	0.612	1224	0.799	0.816	0.017	[-0.025, +0.039]	False
5	3005	0.373	700	0.743	0.787	0.044	[+0.021, +0.087]	True
8	857	0.550	341	0.672	0.741	0.069	[-0.001, +0.126]	False

Table S5: A PREFIX AXIS ON ONE LOG. At prefix k the analyst stands after the case's k-th assignment event: the population is the cases still at risk — more than k events and no group change yet — the target is whether a change occurs AFTER event k, and the baseline is the intake block plus what the prefix has revealed. $k = 0$ is case creation, which is the prediction point every other result in this paper uses, and it is the anchor the rest are read against. Every interval is the pointwise basic construction under the block-weighted bootstrap at 300 draws.

still at risk: more than k events, and no change of assignment group in the first k, because a case whose outcome is already determined by k is not a prediction problem. The **target** is whether a change of group occurs *after* event k, which is the registered handover rule of Section 5 restricted to the remaining suffix, so nothing already observed is predicted. The **baseline** is the intake block plus what the prefix has revealed — the group the case sits in now and how many distinct groups it has seen — which is the prefix-aware baseline that literature builds, and giving the prefix to the baseline and not to the register is the comparison that can embarrass the register. The split is the same fixed temporal holdout under the same declared tie-break, and every interval is the pointwise basic construction under the block-weighted bootstrap at 300 draws. $k = 0$ is case creation, which is the prediction point every other result in this paper occupies.

What it shows. **The register's worth collapses once anything has happened.** At case creation it is +0.257 [+0.233, +0.281] AUC over the intake block on 46,606 cases; two events later it is +0.064 [+0.047, +0.088] on 11,080, and at the deepest prefix measured it retains 27% of its creation-time value. Across the 6 prefixes it ranges over +0.017 to +0.257, and on 2 of them the interval does not exclude zero — the register's sign is not determined by the data at those prefixes at all, though it is determined, and large, at creation.

Two of the three mechanisms are visible in the table. The baseline rises as the prefix reveals the routing history, from 0.641 at creation; the population shrinks and its prevalence falls as the cases that were going to be handed over early leave it; and the register's own cardinality falls with the population. None of this is drift in the register: it is the same field, on the same estate, worth a different amount because the question changed underneath it.

What this does and does not license. It does not make this a predictive-process-monitoring paper, and no claim about that literature's benchmarks is made here: one log, one register, one encoding, no prefix bucketing and no sequence encoder is a pilot and is offered as one. What it does establish is that the prediction point is an argument of the estimand on that literature's own axis and not only on the service-desk one of Section 7.2, and that case creation — the point this paper occupies throughout — is the *most favourable* point for a creation-time register. A study that reported this register's worth from a prefix-based benchmark and a study that reported it from case creation would disagree by a factor of 3.7, and neither would be wrong.

S13. Is the increment stationary across the test half?

Section 11 names two assumptions and says neither is tested: the moving-block construction needs stationarity and mixing that an event log is not guaranteed to have, and the same assumption is what makes Definition 2's population limit well defined under a temporal split. This section tests the first of them, on every pair, and reports the answer whichever way it comes out.

The diagnostic, and the null it is read against. At the reference cell the model is fitted *once* on the training half and then evaluated on the test half cut into 5 consecutive blocks in time. Under a stationary process the per-block increment should differ only by sampling error. A block is a fifth of the test half, so it carries more sampling error than the whole and *some* spread is guaranteed; comparing the spread with the pair's own interval, which is a whole-sample object, would therefore call ordinary noise drift. The reference is instead a **stationarity null**: the same test rows re-partitioned at random into blocks of the same sizes, 400 times, which destroys the time order and nothing else. The observed spread is reported as a percentile of that reference.

log	target	V	range over blocks	spread	null median	percentile	trend	sign varies	drifts
BPIC13_closed	handover	-0.009	[-0.044, +0.011]	0.055	0.054	51.5	0.50	True	False
BPIC13_incidents	duration	0.028	[-0.003, +0.055]	0.058	0.043	76.8	-0.50	True	False
BPIC13_incidents	handover	0.037	[+0.032, +0.042]	0.010	0.022	3.3	-0.50	False	False
BPIC14	duration	0.096	[+0.069, +0.112]	0.043	0.021	98.0	0.10	False	True
BPIC14	handover	0.167	[+0.157, +0.184]	0.027	0.047	13.5	0.00	False	False
BPIC15_1	duration	0.001	[-0.077, +0.103]	0.179	0.069	100.0	-0.10	True	True
BPIC15_1	handover	0.007	[+0.002, +0.012]	0.010	0.021	6.8	0.10	False	False
BPIC15_3	duration	0.035	[-0.013, +0.038]	0.051	0.068	22.5	0.10	True	False
BPIC15_3	handover	-0.003	[-0.053, +0.049]	0.102	0.074	83.5	0.00	True	False
BPIC15_4	duration	0.035	[-0.036, +0.043]	0.079	0.048	92.8	-0.40	True	False
BPIC15_4	handover	0.001	[-0.041, +0.020]	0.062	0.038	92.0	0.67	True	False
BPIC15_5	duration	0.032	[+0.018, +0.107]	0.089	0.106	35.5	0.60	False	False
BPIC17	duration	0.002	[-0.010, +0.011]	0.022	0.015	86.5	-0.60	True	False
BPIC19	duration	0.024	[-0.037, +0.063]	0.100	0.010	100.0	-0.80	True	True
Helpdesk	duration	-0.020	[-0.103, +0.015]	0.118	0.042	100.0	0.30	True	True
RoadFines	duration	0.031	[+0.025, +0.069]	0.044	0.012	100.0	1.00	False	True
Sepsis	duration	0.153	[+0.106, +0.230]	0.124	0.105	64.5	0.60	False	False
UCI498	duration	-0.007	[-0.015, +0.009]	0.025	0.013	99.0	0.10	True	True
UCI498	handover	-0.003	[-0.005, -0.002]	0.003	0.005	9.5	-0.60	False	False

Table S6: IS THE INCREMENT STATIONARY ACROSS THE TEST HALF? The reference cell's model is fitted once on the training half and evaluated on the test half cut into 5 consecutive blocks in time. 'spread' is the range of the per-block increment; 'null median' is the median spread over 400 RANDOM re-partitions of the same rows into blocks of the same sizes, which destroys the time order and nothing else; 'percentile' places the observed spread in that reference. A block is a fifth of the test half and so carries more sampling error than the whole, which is why the comparison is against the null and not against the pair's own interval. 'trend' is the Spearman correlation of the increment with the block index: it separates drift, which has an order, from heterogeneity, which does not. 'drifts' marks a percentile in the null's upper tail at the declared level $\alpha = 0.05$.

What it shows. **On 6 of 19 pairs the increment is not stationary across the test half** — the time-ordered spread sits in the upper tail of the null at the declared level $\alpha = 0.05$ — and on 13 it sits above the null's median. The median observed spread is +0.055 AUC, 1.38 times the null's median, and on the worst pair it is 9.8 times. On 11 of 19 pairs the *sign* of the increment varies between blocks of the same test half.

It is heterogeneity rather than trend. Only 1 of 19 pairs shows a Spearman correlation between the increment and the block index that reaches conventional significance, so what the diagnostic finds is not a register decaying smoothly with time: it is a test half whose periods are not exchangeable. Two candidate mechanisms are in the table. Prevalence moves by a median 0.13 between the first and last block, and by 0.56 on the worst pair; and the share of test rows whose register value was never seen in training rises from a median 4.8% in the first block to 8.2% in the last.

What follows. Two things, and neither is a correction. First, a pair that fails this test is not a pair whose number is wrong: it is a pair whose estimand is a moving target, which is a fact a reader should be given rather than an error to be repaired — and it is the paper's own thesis applied to time, which $\mathcal{S}$ does not contain as an axis. Second, the interval and the band do not cover it. Both are computed under an assumption this diagnostic finds violated on 6 pairs, so on those pairs the interval understates what a reader would see on a different stretch of the same log, and the direction of the understatement is not signed. Adding time as a declared axis — an evaluation period, with the increment reported per period — is the change this measurement implies, and it is not made here.

S14. Results the article summarises

S14.1. The pipeline axis is two axes

Table S8's "learner" column is not a learner axis. Its levels are pipelines: a model family together with an encoding and, in one case, a calibration. On 15 of the 19 pairs only two of them run, so the index is a statement about a two-level axis whose levels differ in two things at once.

On the case study's log the budget allows the two to be crossed. Both model families — penalised logistic regression and histogram gradient boosting — are run against all three encodings: one-hot, frequency, and cross-fitted target encoding, on both of the log's targets. That is 12 complete factorial pipelines, and the decomposition of Section 4.5 then carries a family axis and an encoding axis separately.

Three things follow, and the third is the reason the confound mattered.

The encoding is the larger of the two. It carries 10.1% of the variance in the increment against the family's 5.2%. How the register is turned into columns matters about twice as much as which model consumes them, and it is the choice more often left unstated.

Their interaction is as large as either main effect. The family–encoding two-way component carries 9.7%, the largest two-way term in the crossed decomposition. A gradient-boosting model and a logistic model do not respond to a change of encoding in the same way, which is precisely what an axis that fuses them cannot express.

So the fused index is not the sum of two indices. It is a statement about a two-level axis whose levels move two things at once, and Table S8's pipeline column should be read as that and nothing more. The crossing

log	target	encoding	family	quality level	rung	split	higher-order
BPIC14	duration	0.091	0.066	0.065	0.492	0.014	0.271
BPIC14	handover	0.125	0.049	0.109	0.430	0.009	0.276

Table S7: The decomposition with the model family and the encoding as separate axes, on the case study's log where all six pipelines run. First-order shares, median over the five instruments, under the equal-level measure, then the total higher-order share. The 'learner' axis of Table S8 is the two of them together.

has not been re-run on the other 15 pairs, and Section 11 carries that as a limitation rather than assuming the case study's split transfers.

S14.2. Decision rules, in sample and out

Table S27 compares the four rules of Section 4.7 — *one-number*, *uniform*, *majority* and *weighted mean* — under the loss defined there, inside one instrument at a time and in that instrument's own units. No average in this paper crosses an instrument.

The weighted-mean rule is optimal in sample by construction — its excess regret is exactly zero everywhere, as the lemma requires — so the comparison that matters is out of sample. Choosing the action on a random half of the admissible cells and paying the loss on the other half, the one-number rule costs 0.00450 AUC of expected excess regret against 0.00011 for the majority rule and 0.00002 for the weighted mean. The part of the ordering that holds everywhere is that the one-number rule is worse than the majority rule, in 5 instruments of 5; the full three-way ordering holds in 4 of them.

Three results run against the surface's advocate and are reported as such. The conservative rule — adopt only where the whole surface is positive — is the worst of the four in 4 of 5 instruments, because it declines so often that the performance it forgoes exceeds the performance the one-number rule destroys; the exception is Nagelkerke R^2. The median excess regret of the one-number rule is *zero* in every instrument: on most pairs every rule picks the same action. And under rolling-origin folds — choosing the action on the earlier folds and paying on the later ones, a harder test because the surface itself moves in time — the ordering does not hold: one number costs 0.00885 against the majority rule's 0.01922. Leaving one log out and asking whether a pairwise ordering established on the rest survives on the held-out log, it does on 57% of the 79 comparisons in which the two rules differ at all, which is near chance.

What we conclude from this, and what we do not. We do not conclude that a surface report is a better decision rule than a number. On this corpus the decision-optimal collapse is the weighted mean, that is an algebraic identity in sample and a small and unstable advantage out of it, and on the one utility with an operational meaning the rules mostly tie (Section 8.4). The case for reporting the surface rests on the decomposition and on the sign disagreement, not on the regret comparison, and the regret comparison is reported here as a check on the former rather than as a result in its own right.

S14.3. Calibration, unseen categories and rolling origin

Table S28 reports, for each pair at the reference cell, the calibration intercept and slope, the Brier score, and the share of test cases carrying at least one level unseen in training. The unseen-category rate is the mechanism behind much of the register's behaviour: a register whose identities do not recur across the split cannot help a model that has never seen them, and the rate ranges from 0% to 87%. Calibration is reported because discrimination alone is not a sufficient description of a model [35]; slopes below one indicate the overfitting that high-cardinality one-hot encodings produce, and are the reason a calibrated learner is one of the four.

The frequency-encoded pipeline is the one that is visibly miscalibrated, with a slope well below one, and it is also the one whose increment differs most from the others on the two proper scoring rules. That is the interaction between the pipeline axis and the metric axis, drawn.

Rolling-origin results are in `results/s01_surface.csv` under `split = rolling`*k* and enter the decomposition as the split axis; Supplement S5 prints them per fold. The single holdout is not the best estimate of the increment, and reporting it alone would be a limitation.

S14.4. The case study's leakage and quality experiments

The leakage tipping point. The knowledge reference is load-bearing: it is the field that takes the register's increment to indistinguishable from zero, so the evidence that it was available at τ_2 is stated exactly, and it is weaker than proof. The interaction detail file establishes that the field is written by the service-desk process and not by the incident process — 94,250 interactions in it never became incidents, and 100% of them carry a knowledge reference — but not *when*, for a given incident, the value was written. Table S30 lists each piece of evidence beside what it does and does not establish. Availability at τ_2

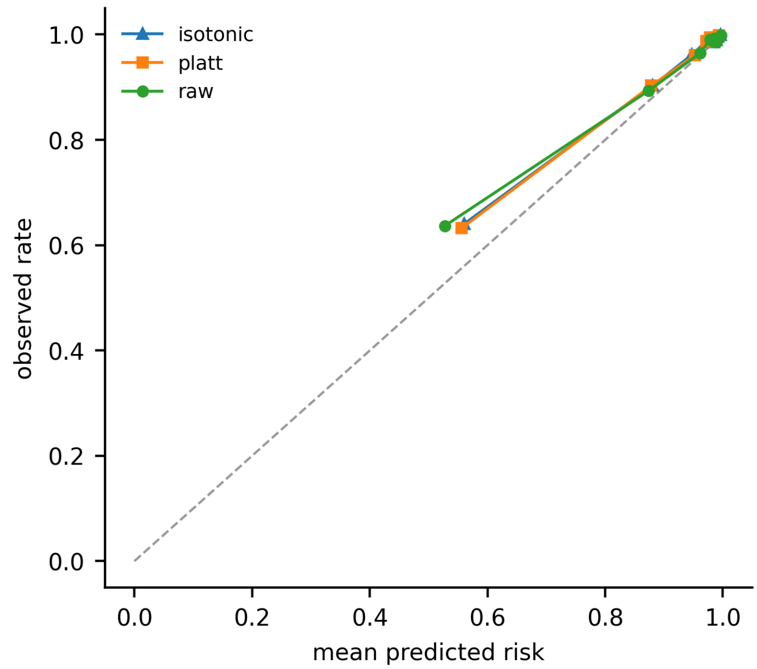

Figure S1: Calibration at the reference cell on the case study's log, by decile of predicted risk. The two proper scoring rules on the metric axis — Brier skill and the log-loss skill — read this picture; the two rank-based ones cannot see it at all (Remark S1). A pipeline whose discrimination is adequate and whose calibration is not therefore moves the increment on some instruments and not others, which is one concrete mechanism behind the metric axis's contribution to Table S8.

is probable and unproven, and the disposition of an unprovable admissibility assumption is a sensitivity analysis rather than a headline.

Because availability is probable rather than proven, the increment is reported in Figure S2 as a function of λ, the share of knowledge references assumed to have been written after the incident existed and therefore inadmissible. Masking a random λ share of the field and re-estimating traces a curve from the permissive end ($\lambda = 0$, the field fully admissible) to the conservative end ($\lambda = 1$, equivalent to τ_2 without the field at all). The masking is stochastic, so each point is a mean over 5 seeds with the spread across them printed beside it, and the curve is a sensitivity display rather than an estimate the paper quotes a number from.

At $\lambda = 0$ the register is worth -0.0029 $[-0.0074, +0.0019]$, which is not resolvably different from zero, and it reaches half of its knowledge-free value at

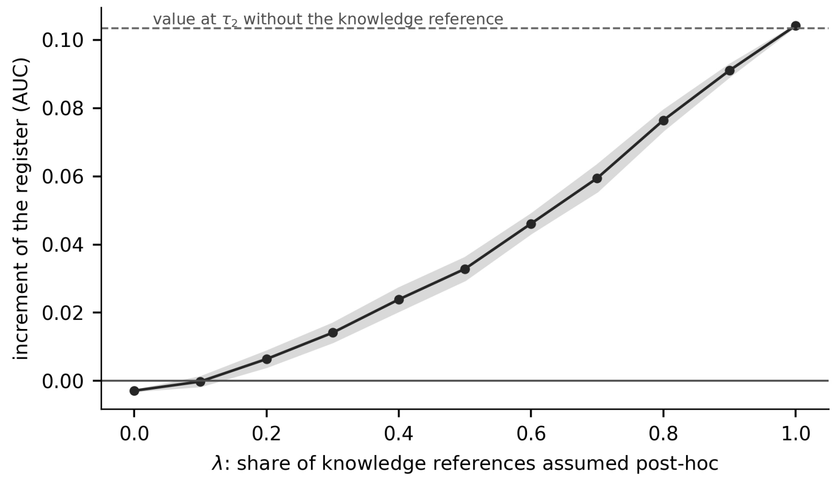

Figure S2: The register's increment at τ_2 on the estate's cohort and reassignment target, as a function of λ, the share of knowledge-article references assumed to have been written after the incident existed. A reader who has a belief about λ can read their own answer off the curve.

$\lambda = 0.64$: a reader who believes that more than 64% of knowledge references are written *after* the incident exists should read this estate's register as worth at least half of +0.1034 AUC, and one who believes almost none are should read it as worth almost nothing. We have no way to measure λ and do not claim one; the curve makes the assumption explicit and continuous rather than binary and made on the reader's behalf.

Register quality. Masking a complete field is a sensitivity experiment, not a model of a real register, so Table S31 reports the six mechanisms of Section 3 plus a discovery lag, and sweeps the accuracy and reconciliation mechanisms rather than reporting one level of each. Everything here is on the estate's cohort and reassignment target; Table S32 carries the severity curves, over 11 severities and 20 seeds per stochastic mechanism, and Table S34 the same mechanisms on the registered cohort. Four things follow, and two of them are nulls we did not expect.

Coverage is the expensive failure, and losing the long tail is not the worst way to lose it. At half population the register is worth +0.0819 if its long tail is absent and +0.0708 if its core is, against +0.1034 complete. Losing the core costs more than losing the tail, which is the opposite of the usual intuition that the rare identities carry the information: on this estate the frequent items are the ones the model has enough cases to learn from. Loss at random is worse than either (+0.0676), because it fragments every identity rather than removing some cleanly.

Accuracy costs about what coverage costs. Corrupting three tenths of the identities — the register is complete, and wrong — leaves it worth +0.0689, a loss of −0.0346, about what losing a third of its population costs. A maturity assessment that counts populated rows and does not sample them for correctness is measuring one of the two and reporting it as the whole.

Reconciliation failure is nearly free. Splitting *every* identity into two records moves the increment by −0.0019: with thousands of cases per identity, halving the evidence behind each still leaves enough.

Staleness is exactly zero, and the reason is the encoder. Removing identities the register had not seen by the split point moves the increment by +0.0000. That is not a bug in the mechanism: a one-hot encoder fitted on the training half already ignores levels it has never seen, so a test row carrying an undiscovered identity contributes an all-zero encoding either way. Under an encoder that keeps missingness as an explicit indicator the effect is non-zero; Supplement S15 measures that on the registered cohort. A *discovery lag* — identities first used in the last half of the training window absent everywhere, leaving 94.0% of rows populated — moves it by +0.0039. Whether a data-quality defect is visible is therefore an interaction between the quality axis and the encoding axis, and an analyst who varies one without the other will conclude that the defect does not matter.

And which rows go missing matters as much as how many. The three *dependent* mechanisms of Section 3 carry severity integrals +0.0768, +0.0486 and +0.0559, against +0.0692 for the independent long-tail mechanism, which the dependent ones bracket rather than track: the identity of the missing rows changes what the loss costs by up to half again, so a quality axis built only from independent coin flips is measuring one corner of the question. The summary is an integral against a uniform measure on severity, and coarsening the sweep from 11 severities to five moves it by at most 0.0114 AUC. Every mechanism is verified by execution to be a function of the training half alone — 16 checks run and 16 pass.

S14.5. Calibration before the decision curve

Why that window, and why the exclusion is not repaired by recalibrating again. A Cox calibration slope of γ means a threshold θ selects the cases a well-calibrated model would select at a different threshold, and the discrepancy grows without bound as γ leaves one; the window is the conventional band in which risks are neither compressed nor stretched beyond a factor the literature treats as readable, and it is a declared convention rather than a derived bound. Widening it to admit everything would not make the excluded thresholds readable as cost ratios; it would only stop saying so. **Recalibration is already an axis, which is why it cannot also be the remedy**: isotonic recalibration is a level of the pipeline axis, so an arm that fails the slope check has, on the pairs that carry it, a recalibrated sibling in the same surface. Applying isotonic recalibration to the failing arm before excluding it would silently replace one specification by another and report the result under the first one's name — the exact substitution this paper's argument is against. The arm is excluded and its sibling is reported as itself. The rule, what it removes and what the curves look like without it are below.

Calibration comes first, or the threshold means nothing. The identity above reads θ as a probability threshold, and everything that follows from it requires the score to be a probability. A threshold of 0.30 applied to a score whose calibration slope is -0.28 is not a statement about a cost ratio of 2.3 false nominations per true one; it is a statement about a number that happens to lie at 0.30 on an arbitrary scale. Every model contributing to a decision-curve conclusion in this paper is therefore refitted under a **training-only** calibration: isotonic and Platt scaling, each cross-validated *inside* the training half, so no test outcome touches the calibrator, and the calibration is rebuilt inside every resampling iteration. Table S29 gives the diagnostics for the raw and the calibrated scores on every pair.

The correction bites. Across the corpus the median calibration slope is 0.63 on raw scores and 0.95 after isotonic calibration; the median expected calibration error falls from 0.105 to 0.089; and the number of arms whose slope lies outside $[0.5, 2]$ falls from 92 to 63. It also changes conclusions: the sign of the net-benefit increment differs between the raw and the calibrated curve at a mean 27.4% of thresholds.

What may and may not be read as a decision curve. Curves computed on calibrated probabilities carry the probability-threshold interpretation and are what Section 8.3 and the case study read. Curves on raw scores are reported as *score-threshold sensitivity analyses* and are labelled so in every table; they say how a conclusion moves as a selection rule tightens, and they do not say what cost ratio that rule encodes.

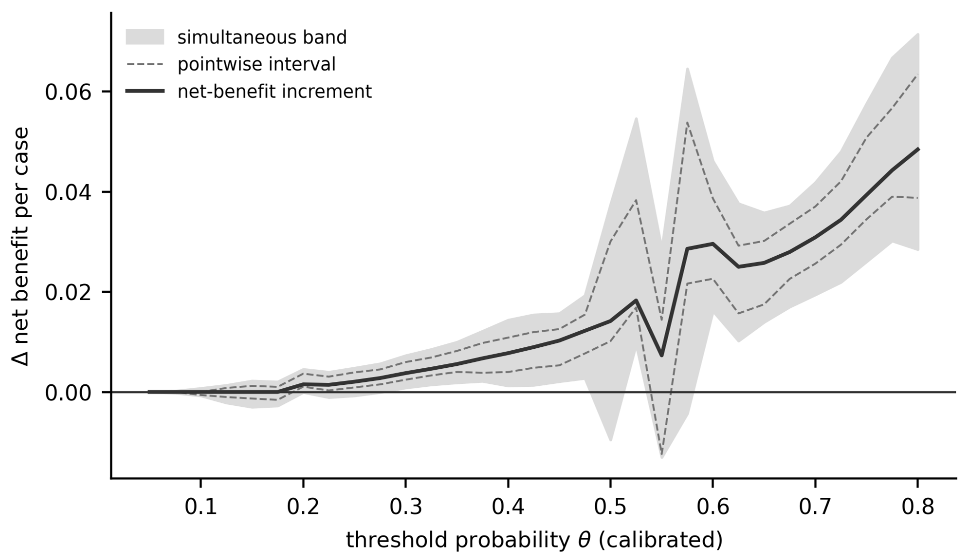

Figure S3: The register's net-benefit increment across the operating range at the reference cell, with the pointwise interval and the simultaneous max-t band. The simultaneous band is what a claim about a *range* of thresholds has to be read against. The curve dips near $\theta = 0.55$, where the simultaneous band's lower edge reaches -0.0130: 0 of the 31 operating points resolve as harmful pointwise and 0 under the simultaneous band, so the dip a reader can see is not a harm the evidence resolves.

S14.6. Simultaneous bands on the decision curve

Figure S3 draws the register's net-benefit increment across the grid at the reference cell on *calibrated* probabilities, with the pointwise interval and the simultaneous max-t band. The band's family is declared separately from the scalar family and spans every admissible net-benefit cell of the pair — 248 members, critical value 3.57 — and the difference from a pointwise reading is not cosmetic. Its median width is 0.0136 against the pointwise interval's 0.0071, and across the 31 operating points of this estate's decision curve the pointwise interval declares the increment resolvably positive at 24 of them while the simultaneous band declares it at 18. Both are built from the same draws, so the only thing that differs is the family: those points are the price of a statement about a *range* of thresholds rather than about each threshold separately, and a paper that reads a decision curve across its range and prices it pointwise has claimed the first while paying for the second.

The curve dips near $\theta = 0.55$, where the simultaneous band's lower edge reaches -0.0130, so the count of points that resolve as *harmful* is reported rather than left to the eye: 0 of the 31 points pointwise and 0 under the simultaneous band. The second can only be smaller than or equal to the first — the wider family gives the wider interval at every point, which `verify_numbers` checks — so a dip a reader can see is not a harm the evidence resolves. Over the 10,974 decision-curve cells the corpus carries, the same counts are 3 and 1; Table S25 prints this pair's own curve, operating point by operating point, and its caption names the cell it is computed at.

And this family is the one most exposed to how the critical value was estimated. The decision-curve bands are *not* coverage-calibrated: the (n, K) calibration of Section 6.3 is estimated for the scalar instruments and is not applied here. On every one of the 19 decision-curve families the multiplier critical value lies below the whole order-statistic interval of the empirical quantile (Section 4.3), by a median factor of 2.09, and across the corpus's decision-curve cells the multiplier band resolves 3,066 against the empirical band's 613. A reader who takes the empirical value should read this section's counts as upper bounds. Section 11 states why we do not choose between the two.

S14.7. The ITSM family, statistic by statistic

Eight of the 19 pairs are IT service management, where the register is a maintained configuration or customer register and a data-quality mechanism has an operational referent. The other eleven include a loan amount treated as a high-cardinality categorical, a case-type reference nobody curates and an article of law, which Section 5.4 names as threats to construct validity. Table S17 reports every headline statistic on all nineteen pairs, on the eight, and on the eleven that remain, each with a cluster-bootstrap interval taking the *log* as the resampling unit — two targets on the same log share a cohort and are not two independent observations.

The higher-order share is 36.5% on the ITSM family against 56.0% on the whole corpus, the largest first-order index 35.5% against 28.7%, and the all-cells sign-disagreement rate 27.1% against 32.8%. One difference runs against the paper. **The headline is lower on the population this paper is about**: of the resolved cells, 3.5% disagree in sign on the ITSM family against 22.2% on the other eleven and 7.5% pooled. The corpus-wide figure is therefore driven by the pairs Section 5.4 names as carrying construct-validity threats, and a reader who takes this paper to be about maintained registers should read the ITSM column and should read it as the weaker case for the apparatus, not the stronger one.

S14.8. The decision rules in full

The four rules of Section 4.7 — *one-number*, *uniform*, *majority* and *weighted mean* — are compared under the loss defined there, inside one instrument at a time and in that instrument's own units. Table S27 and Supplement S14 carry it in full. The weighted mean is optimal in sample by construction, so what matters is out of sample: choosing on a random half of the admissible cells and paying on the other half, the one-number rule costs 0.00450 AUC of expected excess regret against 0.00011 for the majority rule and 0.00002 for the weighted mean, and is worse than the majority rule in 5 instruments of 5.

Three results run against the surface's advocate, and we report them rather than the comparison alone. The conservative rule is the worst of the four in 4 instruments, declining so often that the performance it forgoes exceeds the performance the one-number rule destroys; the median excess regret of the one-number rule is *zero* in every instrument, because on most pairs every rule picks the same action; and under rolling-origin folds the ordering does not hold, at 0.00885 against 0.01922. We therefore do not conclude that a surface report is a better decision rule than a number. The case for reporting the surface rests on the decomposition and on the sign disagreement; this comparison is a check on those and not a result in its own right.

S14.9. The desk's decision, in net benefit per thousand

Because net benefit is a single cardinal utility, the decision-rule comparison of Section 6.6 can be run on it without the unit problem that forces that section to work instrument by instrument. Three things have to be settled first, and each is settled by a declared rule rather than a default.

Which models may enter. A threshold is a cost ratio only if the score is a probability, so a model whose recalibrated slope lies outside $[0.5, 2.0]$ is excluded from the decision-analytic reading by a registered rule applied at the level of the (pair, pipeline, baseline) triple and requiring both arms to pass. A *model* here is one such triple, not one fitted classifier. The rule removes 40 of 118 of them and leaves 3 of 19 pairs with none. **That count is itself the strongest operational finding in this section**: on a third of the models a published evaluation would fit here, no threshold can be read as a cost ratio at all.

Which thresholds. A uniform measure over an operating range from 0.05 to 0.80 is not a desk's belief. For a decision about pre-empting a reassignment the false positive costs a specialist's few minutes and the false negative costs a re-triage, a second queue wait and the service-level exposure that goes with it; ratios between three and ten to one put the indifference threshold between 0.08 and 0.36, with a median of 0.19. A Beta distribution with that support is declared as the *desk distribution*, and single exchange rates are reported beside it.

Which unit. Net benefit per case is a rate near zero, and three decimal places of it look like a tie, so Table S23 reports it per thousand cases.

And the rules still mostly tie. Under the desk distribution the one-number rule, the majority rule and the weighted mean all carry a median excess regret of 0.000 per thousand cases; only the conservative rule loses, at 0.048. The four rules do not all choose the same action on 11 pairs, and where they differ the one-number rule is the worst on 10 of them, at a maximum of 0.412 per thousand — a real ordering and a small quantity. **The case against a one-number report does not rest on this design.** It rests on the decomposition of Section 4.5 and on the sign disagreement of Section 6.4, which is unit-free.

What does separate, in this unit, is the decision time. The register is worth 1.71 true positives per thousand cases against the intake block. Against intake plus group plus the knowledge reference — the same estate, the same register, a later moment in the same process — it is worth 0.003. A desk that read the first number and deployed where the second applies is short 1.70 per thousand cases. That is the operational content of Section 7.2, and it is larger than every difference between the four collapse rules put together. It is also conditional on the knowledge reference being admissible at τ_2, which Section 7 shows is probable and unproven: the second figure is the $\lambda = 0$ end of Figure S2, and a reader who believes a share λ of those references are written after the incident exists should read their own figure off that curve.

What the operational statement does not say. Net benefit converts a discrimination increment into units the desk recognises. It does not convert it into money, and three gaps remain, each needing data this study does not have: nominating a case for review is not the same as preventing a reassignment, and whether acting on a nomination changes the outcome is an interventional question on observational evidence; the exchange rate is declared rather than measured, which is why the whole curve is reported; and the cost of the register itself — discovery tooling, reconciliation effort, the staff who keep it current — is nowhere in this evidence.

S14.10. The measure, the target axis, and which axis leads

The measure is not neutral, and one conclusion does not survive it. Under the reference-weighted measure the median higher-order share is 49.6% and under the concentrated measure 31.7% (Table S26). That second number is smaller than the largest first-order index under the same measure, 39.2%, and on 12 of 19 pairs some single axis carries more variance than everything higher-order combined — against 6 pairs under the equal-level measure. The statement *the interactions carry more than any main effect* is therefore true under the two measures that let a reader roam and false under the one that keeps them near the reference cell. Both invariance tests pass, to 5.6×10^{-17} and 0.004.

The target is an axis too, and it is a small one within a pair. Every index above is computed *within* a log–target pair. On the 6 logs carrying both registered targets the target axis can be added, on the headroom scale, and it carries a median 2.1%. The two targets are different questions, agreeing on a median 51% of cases, so the reading is not that the target hardly matters — on the case study it decides the sign (Section 7.1) — but that once it is fixed, the remaining choices move the answer by more than switching it does.

Which axis leads is a property of the pair. The largest single first-order index anywhere on the corpus is 77.0%, carried by the baseline rung on UCI498 / duration; 4 different axes lead somewhere, and the most frequent leader is the split, on 51% of pair–instrument combinations. An analyst can guess the leader and be right about half the time.

S14.11. What the pooled resolved rate is a ratio of

It is a pooled ratio, it depends on which band resolves the cells, and both facts belong beside it. The denominator is a sum over pairs whose resolved-cell counts run from zero to 332, not an average of per-pair rates, and it depends on which band decides what "resolved" means: the same quantity over the cells the *nominal* multiplier band resolves is 11.9%. The calibrated figure this paper quotes is the smaller of the two, so the choice is the conservative one — but it is a choice, and a reader is entitled to both. Under the empirical critical value of Section 4.3 the denominator is smaller again, 995 cells against 1,150. **And the disagreements are concentrated**: of the 67 cells, 65 come from three logs, and only 4 of the 13 logs contribute any at all. Two of the three are also the pairs with the smallest bootstrap budget (Section 11). On 8 pairs no resolved cell contradicts the reference, on 5 some do, and on 6 nothing resolves.

S14.12. What the case study says to a configuration management programme

S14.13. What this says to a configuration management programme

The register in this log is layered, and the layers are the ones a common service data model uses: an item belongs to a class, a class to a subtype, and a subtype rolls up to a service component. Supplement S4 estimates the increment at each layer against the intake block, on the registered targets. The coarse layers are *not* free — on the handover target the item's type alone is worth +0.134 and its subtype +0.163, against +0.257 for the item itself, and on duration +0.045 against +0.129 — so knowing that a case concerns *a laptop* is worth between a third and two thirds of knowing which one. What the item adds *on top of* the subtype is the quantity a populated hierarchy does not buy, and it is +0.109 on handover and +0.087 on duration. A programme that populates the class hierarchy and stops has bought a real part of the value and left the larger marginal part on the table; it has not bought nothing. These figures are on the registered targets rather than this section's reassignment target, and Supplement S4 says so.

The three dependent quality mechanisms map onto how a configuration management database actually decays, which is why they are in the design space rather than independent coin flips. *Discovery lag* is the interval between a service going live and the discovery tooling finding its items, which is why a new service's cases carry an empty affected item for weeks. *Unreconciled duplicates* arise when two discovery sources write the same physical item under two identifiers. *Propensity-dependent missingness* arises when the affected item is populated by the desk rather than by discovery: the unusual case, which is the one worth predicting, is the one the desk is least likely to classify correctly under time pressure. The measured costs above say which of the three is worth spending on: on this estate, coverage and accuracy at the item level, and not reconciliation.

Two disclaimers. This is predictive performance on one estate for one decision, not procurement advice, and a register that pays nothing here may be bought for change control, audit or impact analysis, none of which this measures; and the ranking of the mechanisms is a property of this cohort's cardinality and volume.

S14.14. The three decision times, moment by moment

An incident in this estate does not begin as an incident. A caller reaches the service desk, an *interaction* is opened, the desk works it, and if it cannot be closed there an incident is created and routed. Three moments at which a prediction could be made are estimated, and what changes between them is not only which fields are admissible (Table 8).

τ_0**, first touch, every call.** The desk has just picked up the call and does not yet know whether it will escalate. The population is every interaction — 145,478 of them — the register is the interaction's own affected item, with 4,034 values, and the admissible baseline is the interaction's intake block. Against it the register is worth +0.2310 [+0.2204, +0.2618].

τ_1**, first touch, the calls that escalate.** The same moment and the same fields, restricted to the 44,513 calls that did become incidents. The register is worth +0.2013 [+0.1930, +0.2224].

τ_2**, incident creation.** The desk has worked the interaction and escalated it. The population is the 45,455 incidents; the group that logged the incident is known, and so — probably — is the knowledge-article reference the desk consulted. Against the intake block the register is worth +0.1838; against intake plus group, +0.1034 [+0.0949, +0.1295]; against intake plus group plus the knowledge reference, −0.0029 [−0.0074, +0.0019] — the one rung of the five that does not resolve.

The pairwise differences separate the mechanisms that a baseline ladder alone conflates, and separating them properly requires one more row than the three moments give. τ_1 and τ_2 are *not* the same cases: 942 incidents have no interaction record, so τ_2 carries 45,455 cases against τ_1's 44,513. Table 8 therefore also reports τ_2 restricted to τ_1's own population — the same 44,513 cases at the same prevalence 0.402 — which is the row that makes the information step an information step.

Restricting the population from every call to the calls that escalate, holding the information fixed, moves the increment by −0.0297 AUC. Moving from first touch to incident creation *on the matched cases*, changing what is known and nothing else, moves it a further −0.0099. The remaining −0.0076 is what the 942 incidents with no interaction record are worth, and it is a population difference rather than an information one: the unadjusted τ_1-to-τ_2

step, -0.0175, is the sum of the two and attributing all of it to information would overstate the information channel by three quarters. Admitting the group and then the knowledge reference takes the increment the rest of the way to zero. A decision time is therefore an argument of the estimand in its own right: it fixes the population, the feature snapshot and the admissible set, and only the third of those is a baseline.

Neither τ_0 nor τ_2 is "the" answer: which one a reader wants depends on where in their own process they intend to predict. A desk screening at τ_0 looks at 145,478 calls a year at a prevalence of 0.146; a desk screening at τ_2 looks at 45,455 incidents at 0.400, and Section 8.4 prices the difference.

S14.15. Magnitude weighting, and the rate under each measure

Weighting by magnitude does not rescue the conventional report, and the rate depends on the measure. Weighting each cell by $|V|$ as well as by the measure gives a median 50.2%, *higher* than the unweighted rate: the cells that disagree are not the negligible ones. The median is 32.8% under the equal-level measure, 25.4% under the reference-weighted and 13.4% under the concentrated, falling as the measure concentrates on the reference specification, which is what it should do. A sign comparison is unit-free, which is why this one quantity may be pooled across instruments where an expected loss may not.

Table S18 attributes the disagreement to axes. Holding everything else fixed and changing one axis, the sign of the increment flips for a median 33.5% of same-everything-else pairs of cells under the split, 31.3% under the pipeline, 22.2% under the baseline rung, 20.4% under the register-quality condition and 19.7% under the metric. The two axes an evaluation paper is most likely to hold fixed without comment — which split, which pipeline — are the two that flip the sign most often.

S14.16. The eight pairs where the construct is what this paper is about

Every corpus statistic is reported on the 8 ITSM-like pairs as well as on all 19 (Table S17), because the register there is a maintained one and the construct is what the paper is about. **The headline is lower on the population this paper is about**: the sign-disagreement rate is 27.1% against 32.8% over the corpus, and on resolved cells 3.5% against 22.2% on the rest. The decomposition barely moves — higher-order 36.5% against 56.0%, largest first-order 35.5% against 28.7%. Supplement S14.7 carries every statistic on both populations.

S14.17. Why no surface is uniformly beneficial, and what that is worth

No surface in this corpus is uniformly beneficial, and that statement is weaker than it looks. It holds under the calibrated band, the nominal one and the empirical one — but *uniformly beneficial* requires every admissible cell's lower band to be above zero, and a cell whose point estimate is negative can never qualify at any critical value. Every one of the 19 pairs contains at least one such cell, so the label is unreachable here before any band, bootstrap or calibration is applied, and the count of zero is not an inferential finding. The statement that carries content needs none of the apparatus: on every pair at least one admissible specification has a negative increment, which is visible from the all-cells column of Table S10. What the apparatus earns is ρ and the resolved counts, not this. For the same reason $\rho = 1$ is not an attainable state for any pair in this design, and Section 4.6's remark that it is the only safe state for a single number should be read as a definition rather than as a hypothesis this corpus tests.

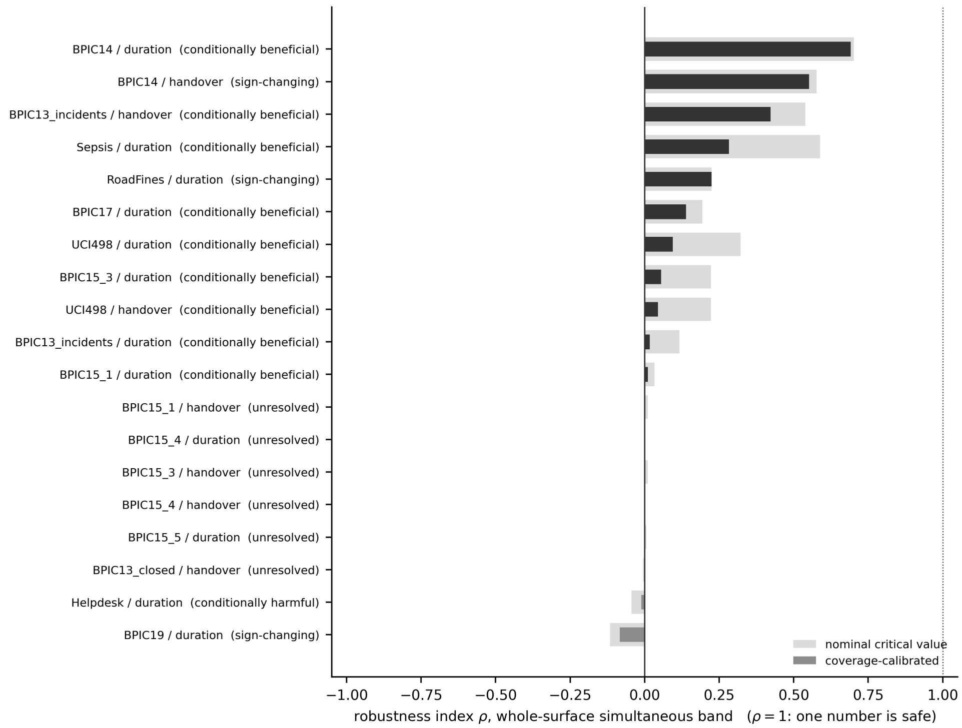

Figure S4: The robustness index ρ per log–target pair, with the region label, under the coverage-calibrated critical value. $\rho = 1$, the dotted line, is the only state in which a single positive number is a safe summary.

S15. Tables the article summarises

The article prints the seven tables a reader needs to follow its argument. This section carries the ones it summarises: the exclusion ledger, the construct audit, the surface denominator, the planned contrasts, the region labels, the ITSM family split, the sign-flip attribution, the cohort decomposition, the axis-kind partition, the specification-curve comparison, the desk utility, the tipping curve, the decision-curve corpus, the declared measures, the four decision rules, the calibration diagnostics, the availability evidence, the severity sweeps, the absorption ladders, and the simulation in full. Every one is generated by `scripts/make_numbers.py` from the result files, and none is typed by hand.

log	target	learner	quality	rung	split	higher-order total	largest involvement
BPIC13_closed	handover	0.110	0.029	0.075	0.203	0.560	0.437
BPIC13_incidents	duration	0.036	0.029	0.029	0.384	0.409	0.372
BPIC13_incidents	handover	0.024	0.099	0.169	0.172	0.308	0.251
BPIC14	duration	0.242	0.151	0.359	0.013	0.233	0.180
BPIC14	handover	0.314	0.127	0.307	0.010	0.233	0.183
BPIC15_1	duration	0.041	0.039	0.055	0.263	0.614	0.548
BPIC15_1	handover	0.018	0.036	0.075	0.079	0.575	0.476
BPIC15_3	duration	0.003	0.081	0.025	0.287	0.607	0.490
BPIC15_3	handover	0.002	0.004	0.128	0.189	0.662	0.547
BPIC15_4	duration	0.060	0.025	0.011	0.111	0.732	0.655
BPIC15_4	handover	0.022	0.031	0.035	0.050	0.793	0.643
BPIC15_5	duration	0.078	0.042	0.057	0.036	0.749	0.664
BPIC17	duration	0.000	0.062	0.001	0.346	0.590	0.494
BPIC19	duration	0.014	0.011	0.003	0.491	0.474	0.418
Helpdesk	duration	0.262	0.086	0.008	0.222	0.339	0.314
RoadFines	duration	0.009	0.289	0.032	0.192	0.449	0.424
Sepsis	duration	0.021	0.145	0.008	0.134	0.608	0.574
UCI498	duration	0.048	0.030	0.056	0.399	0.433	0.391
UCI498	handover	0.079	0.071	0.287	0.136	0.390	0.285

Table S8: The corrected sensitivity decomposition, within instrument, under the equal-level measure. The four first-order indices, then the TOTAL higher-order share $1 - \sum_i S_i$ of Equation (4), then the largest per-axis interaction involvement $\max_i(S_{T_i} - S_i)$ — which is the quantity an earlier version reported as the interaction share and which is not it, because those differences overlap across axes.

S15.1. The corpus

S15.2. Regions, sign flips and the cohort decomposition

S15.3. Measures, rules and calibration

S15.4. The case study

S15.5. The same objects on the registered cohort

Section 7.1 shows that this log gives two answers, and that the difference is the target rather than the cohort. Section 7 prints the estate's own cohort and reassignment target throughout. These are the same objects on the registered cohort and the handover target, so that a reader can see both without either being the one the article quotes.

S15.6. The simulation in full

log	target	K / n	q upper	c	q calibrated	cells (nominal)	rho	cells (cal)	rho (cal)	region
BPIC13_closed	handover	0.324	3.359	1.767	5.935	0 / 1 / 179	-0.006	0 / 0 / 180	0.000	unresolved
BPIC13_incidents	duration	0.133	3.374	1.456	4.914	21 / 0 / 159	0.117	3 / 0 / 177	0.017	conditionally beneficial
BPIC13_incidents	handover	0.133	3.348	1.456	4.876	97 / 0 / 83	0.539	76 / 0 / 104	0.422	conditionally beneficial
BPIC14	duration	0.093	3.509	1.294	4.540	337 / 0 / 143	0.702	332 / 0 / 148	0.692	conditionally beneficial
BPIC14	handover	0.093	3.589	1.294	4.643	279 / 2 / 199	0.577	267 / 2 / 211	0.552	sign-changing
BPIC15_1	duration	0.106	3.357	1.343	4.509	6 / 0 / 174	0.033	2 / 0 / 178	0.011	conditionally beneficial
BPIC15_1	handover	0.106	3.382	1.343	4.542	2 / 0 / 178	0.011	0 / 0 / 180	0.000	unresolved
BPIC15_3	duration	0.095	3.353	1.300	4.358	40 / 0 / 140	0.222	10 / 0 / 170	0.056	conditionally beneficial
BPIC15_3	handover	0.095	3.339	1.300	4.341	2 / 0 / 178	0.011	0 / 0 / 180	0.000	unresolved
BPIC15_4	duration	0.072	3.400	1.243	4.227	0 / 0 / 180	0.000	0 / 0 / 180	0.000	unresolved
BPIC15_4	handover	0.072	3.447	1.243	4.286	0 / 0 / 180	0.000	0 / 0 / 180	0.000	unresolved
BPIC15_5	duration	0.132	3.412	1.455	4.964	4 / 3 / 173	0.006	0 / 0 / 180	0.000	unresolved
BPIC17	duration	0.032	3.271	1.177	3.849	35 / 0 / 145	0.194	25 / 0 / 155	0.139	conditionally beneficial
BPIC19	duration	0.011	3.336	1.177	3.926	14 / 28 / 78	-0.117	14 / 24 / 82	-0.083	sign-changing
Helpdesk	duration	0.123	3.333	1.431	4.769	0 / 8 / 172	-0.044	0 / 2 / 178	-0.011	conditionally harmful
RoadFines	duration	0.001	3.158	1.177	3.717	47 / 20 / 53	0.225	45 / 18 / 57	0.225	sign-changing
Sepsis	duration	0.200	3.338	1.562	5.215	106 / 0 / 74	0.589	51 / 0 / 129	0.283	conditionally beneficial
UCI498	duration	0.301	3.276	1.756	5.751	58 / 0 / 122	0.322	17 / 0 / 163	0.094	conditionally beneficial
UCI498	handover	0.301	3.349	1.756	5.880	40 / 0 / 140	0.222	8 / 0 / 172	0.044	conditionally beneficial

Table S9: Resolution regions under the nominal critical value and under one calibrated for the register's cardinality. c is the inflation factor the (n, K) plane of Section 10.3 gives at that pair's own K/n; the calibrated critical value is c times the nominal one. 'q upper' is the UPPER end of the critical value's Monte Carlo interval, which is the value the bands use by the rule of Section 4.3 that a cell resolves only at the conservative end; the point estimate whose corpus median the text quotes is smaller. The cell columns are beneficial / harmful / unresolved, and the admissible count is the same under both critical values. The calibrated columns are the ones the article's counts use; the nominal ones are printed beside them so that the size of the correction is visible.

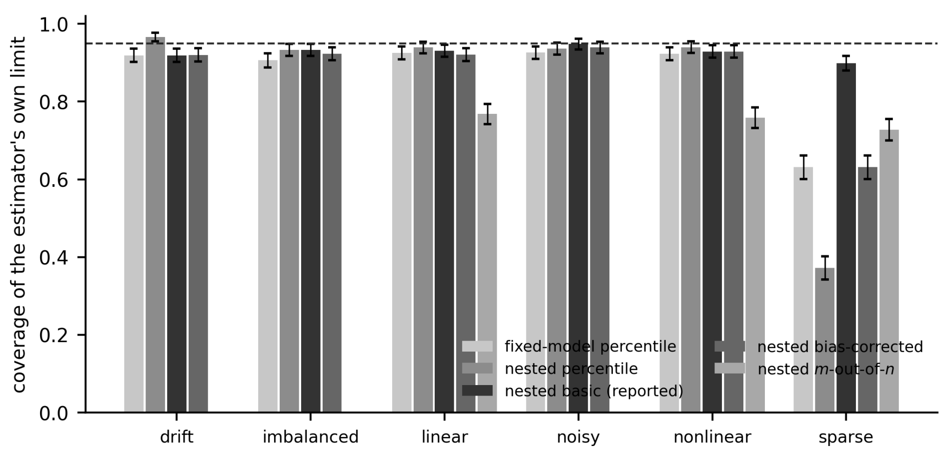

Figure S5: Coverage of the estimator's own limit in six simulated worlds, by interval construction, with Monte Carlo error bars. The dashed line is nominal. The fixed-model percentile interval is the conventional one; the nested basic interval is the one this paper reports. The sparse world is where they disagree by more than a few points, and the nested percentile bar there is the failure the basic construction repairs.

log	target	cells	all-cells rate	resolved / of which disagree	resolved rate	magnitude rate
BPIC13_closed	handover	1080	0.284	0 / 0	–	0.111
BPIC13_incidents	duration	1080	0.393	3 / 0	0.000	0.812
BPIC13_incidents	handover	1080	0.111	76 / 0	0.000	0.482
BPIC14	duration	4800	0.106	332 / 0	0.000	0.018
BPIC14	handover	4800	0.244	269 / 2	0.007	0.025
BPIC15_1	duration	1080	0.423	2 / 0	0.000	0.502
BPIC15_1	handover	1080	0.242	0 / 0	–	0.529
BPIC15_3	duration	1080	0.191	10 / 0	0.000	0.179
BPIC15_3	handover	1080	0.616	0 / 0	–	0.565
BPIC15_4	duration	1080	0.623	0 / 0	–	0.715
BPIC15_4	handover	1080	0.618	0 / 0	–	0.784
BPIC15_5	duration	1080	0.328	0 / 0	–	0.507
BPIC17	duration	1080	0.353	25 / 0	0.000	0.203
BPIC19	duration	1620	0.360	38 / 24	0.632	0.563
Helpdesk	duration	1080	0.257	2 / 0	0.000	0.102
RoadFines	duration	1620	0.173	63 / 18	0.286	0.099
Sepsis	duration	1080	0.056	51 / 0	0.000	0.031
UCI498	duration	1080	0.493	17 / 17	1.000	0.623
UCI498	handover	1080	0.481	8 / 6	0.750	0.683

Table S10: The sign-disagreement rate of a one-number report, counted three ways, under the equal-level measure. The all-cells rate counts every admissible cell, most of whose signs the data do not determine; the resolved rate is restricted to the cells whose coverage-calibrated whole-surface band excludes zero, and the column before it gives the two counts it is computed from; the magnitude rate weights every cell by $|V|$ as well as by the measure. A pair with no resolved cell has no resolved rate and is printed with none rather than with zero.

step	cohort	target	V at the knowledge rung	lo	hi	delta
the case study as published	reassignment	reassignment	-0.00322	–	–	–
the case study, tie-break declared	reassignment	reassignment	-0.00286	-0.00730	0.00203	0.00035
then the other tie-break rule	reassignment	reassignment	-0.00286	-0.00730	0.00203	0.00000
then the registered cohort	registered	reassignment	-0.00385	-0.00898	0.00069	-0.00098
then the registered target	registered	handover	0.00768	-0.00161	0.01217	0.01153

Table S11: The two published values of the register's increment at the later decision time, reconciled one factor at a time. Each row changes exactly one thing from the row above; 'delta' is what that change is worth. The first row is the number Section 7 reported and the last is the planned contrast PC3 of Table S15. Intervals are the pointwise basic construction at 400 draws; the first row has none because the ordering it was computed under is not reproducible by a rule, which is the subject of the second row.

log	domain	code	detail
Helpdesk	itsm	NO_HEADROOM	handover/primary prev=0.9993
Helpdesk	itsm	NO_HEADROOM	handover/g=workgroup prev=0.0011
BPIC13_open	itsm	SMALL_N	n=819
BPIC13_closed	itsm	NO_TEST_VARIATION	duration
BPIC12	lending	NO_G	n=13087
BPIC17	lending	NO_HEADROOM	handover/primary prev=0.9977
BPIC19	procurement	NO_HEADROOM	handover/primary prev=0.9631
BPIC15_2	permitting	SMALL_N	n=832
BPIC15_5	permitting	NO_HEADROOM	handover/primary prev=0.9593
BPIC20_domestic	expenses	NO_G	n=10500
BPIC20_intl	expenses	NO_G	n=6449
BPIC20_prepaid	expenses	NO_G	n=2099
BPIC20_permit	expenses	NO_G	n=7065
BPIC20_rfp	expenses	NO_G	n=6886
Sepsis	healthcare	NO_HEADROOM	handover/primary prev=1.0000
HospitalBilling	healthcare	NO_F	n=100000
RoadFines	enforcement	NO_HEADROOM	handover/primary prev=0.0037
BPIC11	healthcare	HELD_OUT	downloaded as a held-out log for the out-of-corpus test; never offered to the admission rules
BPIC18	agriculture	NO_TEST_VARIATION	held-out set: duration
BPIC18	agriculture	NO_HEADROOM	held-out set: handover prev=1.0000

Table S12: Every downloaded log that is not among the 19 admitted log–target pairs, with its registered exclusion code. HELD_OUT names the logs downloaded as a held-out set for a separate test and never offered to the admission rules; every other code is a rule of Section 5. Some logs appear here under one target and among the admitted pairs under the other, which is what a per-pair admission rule does.

log	domain	target	register	free field	why it behaves as a register	threat to construct validity
RoadFines	enforcement	duration	article	org:resource	article of law, reused across fines	semantics legal, not operational
Sepsis	healthcare	duration	Diagnose	org:group	diagnosis vocabulary: reused, not maintained	no maintenance process; quality axis hypothetical
BPIC13_closed	itsm	handover	product	org:resource	maintained CI or customer register	none beyond domain: closest to the case study
BPIC13_incidents	itsm	duration	product	org:resource	maintained CI or customer register	none beyond domain: closest to the case study
BPIC13_incidents	itsm	handover	product	org:resource	maintained CI or customer register	none beyond domain: closest to the case study
BPIC14	itsm	duration	CI Name (aff)	assignment group	maintained CI or customer register	none beyond domain: closest to the case study
BPIC14	itsm	handover	CI Name (aff)	assignment group	maintained CI or customer register	none beyond domain: closest to the case study
Helpdesk	itsm	duration	customer	Resource	maintained CI or customer register	none beyond domain: closest to the case study
UCI498	itsm	duration	caller_id	assigned_to	maintained CI or customer register	none beyond domain: closest to the case study
UCI498	itsm	handover	caller_id	assigned_to	maintained CI or customer register	none beyond domain: closest to the case study
BPIC17	lending	duration	case:RequestedAmount	org:resource	application or offer reference	near-unique per case, so reuse is thin
BPIC15_1	permitting	duration	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_1	permitting	handover	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_3	permitting	duration	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_3	permitting	handover	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_4	permitting	duration	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_4	permitting	handover	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC15_5	permitting	duration	case:parts	org:resource	case-type reference, reused	nobody curates it
BPIC19	procurement	duration	case:Vendor	org:resource	item or vendor catalogue entry	target is a duration, not a routing decision

Table S13: Construct validity, per log–target pair: the field playing the register role, why it behaves as a maintained and reused entity, the free field, and the specific threat to construct validity the pair carries. These are not the same construct, which is why the corpus is a benchmark of specification sensitivity rather than a replication.

log	target	learners	splits	quality	rungs	metrics	declared scalar	computed scalar	model fits	duplicates	missing	inference cells	draws	register levels K	K / n train
BPIC13_closed	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	337	0.324
BPIC13_incidents	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	704	0.133
BPIC13_incidents	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	704	0.133
BPIC14	duration	4	6	10	4	5	4800	4800	1200	0	0	96	150	3019	0.093
BPIC14	handover	4	6	10	4	5	4800	4800	1200	0	0	96	150	3019	0.093
BPIC15_1	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	89	0.106
BPIC15_1	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	89	0.106
BPIC15_3	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	94	0.095
BPIC15_3	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	94	0.095
BPIC15_4	duration	2	6	6	3	5	1080	1080	288	0	0	36	150	53	0.072
BPIC15_4	handover	2	6	6	3	5	1080	1080	288	0	0	36	150	53	0.072
BPIC15_5	duration	2	6	6	3	5	1080	1080	288	0	0	36	150	107	0.132
BPIC17	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	701	0.032
BPIC19	duration	3	6	6	3	5	1620	1620	432	0	0	24	40	1975	0.011
Helpdesk	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	394	0.123
RoadFines	duration	3	6	6	3	5	1620	1620	432	0	0	24	40	66	0.001
Sepsis	duration	2	6	6	3	5	1080	1080	288	0	0	36	150	147	0.200
UCI498	duration	2	6	6	3	5	1080	1080	288	0	0	36	80	5245	0.301
UCI498	handover	2	6	6	3	5	1080	1080	288	0	0	36	80	5245	0.301

Table S14: The surface denominator audit. For every log–target pair: the declared level count on each axis, then two counts *on the same basis* — the admissible scalar cells the declaration multiplies out, and the same cells counted in the surface file — which are equal on every pair. 'model fits' is a different quantity and is printed separately: it counts fitted arms before the expansion over instruments and *includes* the intercept-only rung, which the master surface carries so that the surface is complete and which no admissible set contains. Then duplicates, missing combinations, the size and draw count of the inference surface, the register's cardinality and its ratio to the training row count, which Section 10.3 shows is what governs whether the bands cover at their nominal level.

contrast	cell a	cell b	estimate	interval	p Holm	Holm	resolved
PC1	intake	–	0.25718	[+0.23556, +0.27809]	0.00250	True	True
PC2	intake + g	–	0.16727	[+0.14456, +0.20735]	0.00250	True	True
PC3	intake + g + km	–	0.00768	[-0.00059, +0.01210]	0.00250	True	False
PC4	intake	intake + g	0.08991	[+0.04844, +0.11706]	0.00250	True	True
PC5	intake + g	intake + g, long tail 50%	0.06339	[+0.05368, +0.07740]	0.00250	True	True

Table S15: The five final-round *planned* contrasts (PC1–PC5), stated in words in Section 4.4. Every one is on the REGISTERED cohort (46,606 cases) and the handover target — not the case study's cohort and reassignment target, which Section 7.1 reconciles against these — at incident creation, with the register clean, under one-hot logistic regression on the single temporal holdout in ROC AUC; PC3 additionally admits the knowledge reference and PC5's second cell carries the quality named in it. p-values are ONE-SIDED bootstrap values from the plus-one estimator $(k+1)/(B+1)$ with a Holm step-down correction; the intervals are TWO-SIDED basic intervals from the same draws, and the last column says whether the interval resolves — which is the criterion this paper uses and is not the same as the p-value's. PC3 is the contrast on which they differ, and Section 4.4 derives why. The contrasts are not confirmatory: they were fixed before this analysis round, not before the data were seen.

log	target	cells	beneficial	harmful	unresolved	rho	region	q	label under the narrow family
BPIC13_closed	handover	180	0	1	179	-0.006	conditionally harmful	3.338	sign-changing
BPIC13_incidents	duration	180	21	0	159	0.117	conditionally beneficial	3.354	sign-changing
BPIC13_incidents	handover	180	97	0	83	0.539	conditionally beneficial	3.330	conditionally beneficial
BPIC14	duration	480	337	0	143	0.702	conditionally beneficial	3.492	conditionally beneficial
BPIC14	handover	480	279	2	199	0.577	sign-changing	3.569	sign-changing
BPIC15_1	duration	180	6	0	174	0.033	conditionally beneficial	3.339	conditionally beneficial
BPIC15_1	handover	180	2	0	178	0.011	conditionally beneficial	3.358	sign-changing
BPIC15_3	duration	180	40	0	140	0.222	conditionally beneficial	3.335	sign-changing
BPIC15_3	handover	180	2	0	178	0.011	conditionally beneficial	3.318	sign-changing
BPIC15_4	duration	180	0	0	180	0.000	unresolved	3.380	sign-changing
BPIC15_4	handover	180	0	0	180	0.000	unresolved	3.425	sign-changing
BPIC15_5	duration	180	4	3	173	0.006	sign-changing	3.395	sign-changing
BPIC17	duration	180	35	0	145	0.194	conditionally beneficial	3.252	conditionally beneficial
BPIC19	duration	120	14	28	78	-0.117	sign-changing	3.317	sign-changing
Helpdesk	duration	180	0	8	172	-0.044	conditionally harmful	3.312	sign-changing
RoadFines	duration	120	47	20	53	0.225	sign-changing	3.140	sign-changing
Sepsis	duration	180	106	0	74	0.589	conditionally beneficial	3.320	conditionally beneficial
UCI498	duration	180	58	0	122	0.322	conditionally beneficial	3.257	conditionally beneficial
UCI498	handover	180	40	0	140	0.222	conditionally beneficial	3.329	sign-changing

Table S16: Resolution regions from the WHOLE-SURFACE simultaneous band at its NOMINAL critical value, with that value q and, for comparison, the label a narrower within-instrument family would have given the same draws. **These are not the labels the article quotes**: the article's regions and ρ are computed under the coverage-calibrated critical value and are in Table S9. This table is printed so that the cost of the wider FAMILY is visible separately from the cost of the CALIBRATION.

statistic	all 19	itsm 8	other 11
baseline spread (AUC)	0.073 [0.024, 0.097]	0.070 [0.014, 0.132]	0.088 [0.022, 0.097]
higher-order share	0.560 [0.409, 0.608]	0.365 [0.233, 0.450]	0.608 [0.532, 0.732]
largest first-order index	0.287 [0.185, 0.352]	0.355 [0.278, 0.358]	0.201 [0.151, 0.305]
misreport, all cells	0.328 [0.243, 0.452]	0.271 [0.244, 0.481]	0.353 [0.191, 0.617]
misreport, analyst latitude only	0.200 [0.100, 0.467]	0.194 [0.067, 0.367]	0.400 [0.067, 0.600]
misreport, magnitude-weighted	0.502 [0.102, 0.623]	0.296 [0.025, 0.623]	0.507 [0.139, 0.640]
misreport, resolved cells (pooled)	0.075	0.035	0.222
robustness index rho (calibrated)	0.017 [0.000, 0.182]	0.069 [-0.006, 0.552]	0.000 [0.000, 0.182]
one-number excess regret (AUC)	0.000 [0.000, 0.003]	0.000 [0.000, 0.011]	0.000 [0.000, 0.005]

Table S17: Every headline corpus statistic on three populations: all nineteen log–target pairs, the eight from IT service management where the register is a maintained configuration or customer register, and the eleven where it is not. Each cell is the statistic with a 95% percentile interval from a cluster bootstrap that resamples LOGS, because two targets on the same log share a cohort and are not two independent observations. The resolved-cell row is a pooled ratio — disagreeing cells over resolved cells, summed across the population — and not a median of per-pair rates, which on a population where most pairs resolve little prints zero and says nothing.

log	target	learner	metric	quality	rung	split
BPIC13_closed	handover	0.287	0.214	0.234	0.222	0.401
BPIC13_incidents	duration	0.374	0.249	0.204	0.230	0.405
BPIC13_incidents	handover	0.174	0.089	0.111	0.100	0.181
BPIC14	duration	0.155	0.051	0.121	0.150	0.089
BPIC14	handover	0.426	0.096	0.115	0.139	0.113
BPIC15_1	duration	0.383	0.217	0.228	0.348	0.446
BPIC15_1	handover	0.313	0.168	0.232	0.263	0.331
BPIC15_3	duration	0.337	0.144	0.164	0.187	0.250
BPIC15_3	handover	0.365	0.302	0.217	0.276	0.449
BPIC15_4	duration	0.431	0.169	0.238	0.196	0.501
BPIC15_4	handover	0.450	0.254	0.255	0.287	0.454
BPIC15_5	duration	0.426	0.194	0.221	0.309	0.335
BPIC17	duration	0.291	0.219	0.257	0.224	0.397
BPIC19	duration	0.183	0.217	0.160	0.079	0.451
Helpdesk	duration	0.252	0.273	0.181	0.131	0.284
RoadFines	duration	0.120	0.172	0.181	0.112	0.221
Sepsis	duration	0.113	0.066	0.058	0.063	0.089
UCI498	duration	0.300	0.197	0.199	0.383	0.384
UCI498	handover	0.333	0.206	0.230	0.489	0.289

Table S18: Sign flips attributable to each axis: the share of pairs of cells that agree on every other axis and disagree on the sign of the increment.

effect	B intake	B intake g	B intake g km
cohort	0.000	0.000	0.000
cohort x target	0.000	0.000	0.007
target	1.000	1.000	0.993

Table S19: The exact two-factor decomposition of the discrepancy. Share of the variance of V over the cohort $\times$ target design, at each rung of the ladder, at a declared tie-break. With two levels per factor the decomposition is exact.

log	target	lat. raw	res. raw	ctf. raw	h.o. raw	lat. hdrm	res. hdrm	ctf. hdrm	h.o. hdrm
BPIC13_closed	handover	0.223	0.203	0.029	0.546	0.192	0.111	0.014	0.683
BPIC13_incidents	duration	0.111	0.384	0.029	0.475	0.101	0.158	0.011	0.729
BPIC13_incidents	handover	0.174	0.172	0.099	0.555	0.080	0.094	0.031	0.795
BPIC14	duration	0.606	0.013	0.151	0.231	0.641	0.008	0.104	0.247
BPIC14	handover	0.631	0.010	0.127	0.232	0.612	0.007	0.065	0.316
BPIC15_1	duration	0.084	0.263	0.039	0.613	0.160	0.141	0.005	0.694
BPIC15_1	handover	0.106	0.079	0.036	0.779	0.043	0.060	0.027	0.869
BPIC15_3	duration	0.037	0.287	0.081	0.595	0.107	0.184	0.068	0.641
BPIC15_3	handover	0.128	0.189	0.004	0.679	0.088	0.125	0.001	0.786
BPIC15_4	duration	0.083	0.111	0.025	0.781	0.062	0.156	0.007	0.776
BPIC15_4	handover	0.090	0.050	0.031	0.829	0.089	0.017	0.038	0.856
BPIC15_5	duration	0.157	0.036	0.042	0.766	0.231	0.043	0.020	0.705
BPIC17	duration	0.001	0.346	0.062	0.591	0.408	0.087	0.046	0.459
BPIC19	duration	0.017	0.491	0.011	0.481	0.165	0.374	0.010	0.451
Helpdesk	duration	0.287	0.222	0.086	0.405	0.356	0.151	0.065	0.428
RoadFines	duration	0.065	0.192	0.289	0.454	0.534	0.016	0.090	0.360
Sepsis	duration	0.127	0.134	0.145	0.594	0.366	0.061	0.075	0.498
UCI498	duration	0.107	0.399	0.030	0.465	0.343	0.116	0.008	0.533
UCI498	handover	0.396	0.136	0.071	0.398	0.427	0.057	0.025	0.491

Table S20: The variance decomposition partitioned by what KIND of choice each axis is, under both declared scales. *Analyst latitude* is the pipeline and the baseline rung — and, on the second scale only, the instrument — things an analyst chooses and could have chosen otherwise. *Resampling* is the split, five of whose six levels are expanding-origin folds on the same data and are therefore different samples rather than different analyses. *Counterfactual* is the register-quality condition, which is a different state of the world. The remainder belongs to no single axis. *Columns.* lat. is analyst latitude, res. resampling, ctf. the counterfactual and h.o. the higher-order remainder; raw and hdrm are the two scales. *Which scale, and why there are two.* raw, within instrument is the paper's primary scale: the decomposition is taken separately inside each of the five instruments, on that instrument's own units, and the median is reported — so the instrument is a stratum here and its own contribution appears in no column. headroom, instrument as an axis is the secondary scale of Section 4.5: raw AUC and raw average precision do not share units, so making the instrument a sixth axis requires the headroom rescaling. Every row is the equal-level measure. The two scales order *analyst latitude* against *resampling* oppositely, which is a property of the scale and not of the corpus; Section S14.16 says so rather than choosing one.

axis varied	reference level	pairs	median rate	min over pairs	max over pairs
the declared reference	–	19	0.328	0.056	0.623
pipeline	boosting, target-encoded	19	0.353	0.056	0.716
pipeline	boosting, isotonic	19	0.328	0.056	0.640
pipeline	logistic, frequency-encoded	19	0.353	0.056	0.756
instrument	average precision	19	0.328	0.056	0.623
instrument	Brier skill	19	0.382	0.056	0.672
instrument	log-loss skill	19	0.382	0.056	0.672
instrument	nagelkerke	19	0.382	0.056	0.672
baseline rung	half the intake block	19	0.328	0.056	0.623
baseline rung	intake only	19	0.328	0.056	0.623
baseline rung	intake, group and knowledge	19	0.353	0.056	0.894
split	rolling1	19	0.353	0.056	0.743
split	rolling2	19	0.328	0.056	0.493
split	rolling3	19	0.353	0.056	0.743
split	rolling4	19	0.328	0.056	0.623
split	rolling5	19	0.384	0.056	0.716
register quality	clean at 1.0	19	0.328	0.056	0.623
register quality	corrupt at 0.05	19	0.328	0.056	0.623
register quality	corrupt at 0.15	19	0.328	0.056	0.647
register quality	duplicate at 0.15	19	0.328	0.056	0.623
register quality	mask common at 0.5	19	0.328	0.056	0.623
register quality	mask random at 0.5	19	0.328	0.056	0.623
register quality	mask rare at 0.25	19	0.382	0.056	0.889
register quality	mask rare at 0.5	19	0.328	0.056	0.623
register quality	mask rare at 0.75	19	0.328	0.056	0.623
register quality	stale at 1.0	19	0.328	0.056	0.623

Table S21: The headline sign-disagreement rate under a different declared reference specification. Each row moves the reference ONE axis at a time and holds the rest at the declared cell, then recomputes the all-cells rate on every pair under the equal-level measure; the first row is the paper's own reference. The reference supplies the sign the other cells are counted as agreeing or disagreeing with, so moving it moves which cells disagree and nothing else. The median column is the corpus median the article quotes; the two beside it are over pairs and show that the spread within the corpus is far larger than the spread across reference specifications.

log	target	cases	capped	observed median	null median	p (median)	share positive	SCA verdict	region	rho
BPIC13_closed	handover	1487	False	-0.010	-0.008	0.327	0.208	not distinguishable from null	unresolved	0.000
BPIC13_incidents	duration	7554	False	0.010	-0.012	0.673	0.625	not distinguishable from null	conditionally beneficial	0.017
BPIC13_incidents	handover	7554	False	0.031	-0.006	0.010	0.833	non-null at 5%	conditionally beneficial	0.422
BPIC14	duration	20000	True	0.091	-0.001	0.010	1.000	non-null at 5%	conditionally beneficial	0.692
BPIC14	handover	20000	True	0.035	0.000	0.010	0.750	non-null at 5%	sign-changing	0.552
BPIC15_1	duration	1199	False	0.030	0.001	0.010	0.792	non-null at 5%	conditionally beneficial	0.011
BPIC15_1	handover	1199	False	0.002	-0.002	0.653	0.708	not distinguishable from null	unresolved	0.000
BPIC15_3	duration	1409	False	0.119	0.003	0.010	0.958	non-null at 5%	conditionally beneficial	0.056
BPIC15_3	handover	1409	False	0.023	-0.005	0.020	0.625	non-null at 5%	unresolved	0.000
BPIC15_4	duration	1053	False	-0.030	-0.004	0.010	0.333	non-null at 5%	unresolved	0.000
BPIC15_4	handover	1053	False	-0.018	-0.011	0.158	0.250	not distinguishable from null	unresolved	0.000
BPIC15_5	duration	1156	False	0.043	-0.001	0.010	0.833	non-null at 5%	unresolved	0.000
BPIC17	duration	20000	True	0.006	-0.001	0.030	0.708	non-null at 5%	conditionally beneficial	0.139
BPIC19	duration	20000	True	0.051	-0.001	0.010	1.000	non-null at 5%	sign-changing	-0.083
Helpdesk	duration	4580	False	-0.013	-0.008	0.109	0.208	not distinguishable from null	conditionally harmful	-0.011
RoadFines	duration	20000	True	0.016	0.000	0.010	0.792	non-null at 5%	sign-changing	0.225
Sepsis	duration	1050	False	0.154	-0.003	0.010	1.000	non-null at 5%	conditionally beneficial	0.283
UCI498	duration	20000	True	0.005	-0.005	0.396	0.625	not distinguishable from null	conditionally beneficial	0.094
UCI498	handover	20000	True	0.000	-0.002	1.000	0.500	not distinguishable from null	conditionally beneficial	0.044

Table S22: Specification-curve analysis as practised, beside this paper's region label. The permutation test asks whether the surface as a whole differs from one with no signal in it; the region asks, of each cell, whether the sign of that cell's own increment is determined. Where the two part company is what the extra machinery is for. Both verdicts are computed on the declared 24-cell sub-surface at 100 permutations; 'cases' is the number of cases the permutation test ran on, which is the pair's own count unless the declared cap of 20,000 bound on it. The region column is the coverage-calibrated label of Section 6.3, computed on the full data.

threshold measure	majority	one-number	uniform-beneficial	weighted-mean	pairs where the rules differ
desk distribution	0	0	0.0477	0	11
fixed 1:19	0	0	0.0000	0	5
fixed 1:2	0	0	0.3251	0	10
fixed 1:4	0	0	0.2127	0	11
fixed 1:9	0	0	0.0000	0	7
uniform over the declared range	0	0	1.3468	0	13

Table S23: Median excess regret of each decision rule, in net benefit per thousand cases, on the models that pass the registered calibration rule. The desk distribution places its mass on the exchange rates a service desk plausibly holds; the fixed rows report single exchange rates so that a reader who rejects the distribution still has a number. The last column counts the pairs on which the four rules do not all lose the same.

lambda posthoc	V auc	sd	n seeds
0.000	-0.003	0.000	5
0.100	0.000	0.001	5
0.200	0.006	0.002	5
0.300	0.014	0.003	5
0.400	0.024	0.003	5
0.500	0.033	0.003	5
0.600	0.046	0.003	5
0.700	0.059	0.004	5
0.800	0.076	0.003	5
0.900	0.091	0.002	5
1.000	0.104	0.000	5

Table S24: The register's increment against the share of knowledge references assumed post-hoc.

metric	V	se	sim lo	sim hi	sim resolved
nb_0.050	0.000	0.000	0.000	0.000	False
nb_0.075	0.000	0.000	0.000	0.000	False
nb_0.100	0.000	0.000	0.000	0.000	False
nb_0.125	0.000	0.000	-0.001	0.001	False
nb_0.150	0.001	0.001	-0.002	0.004	False
nb_0.175	0.003	0.001	0.001	0.008	True
nb_0.200	0.003	0.001	0.000	0.007	False
nb_0.225	0.004	0.001	0.001	0.007	True
nb_0.250	0.004	0.001	0.001	0.008	True
nb_0.275	0.005	0.001	0.001	0.008	True
nb_0.300	0.005	0.001	0.001	0.009	True
nb_0.325	0.006	0.001	0.002	0.010	True
nb_0.350	0.007	0.001	0.002	0.012	True
nb_0.375	0.008	0.001	0.002	0.013	True
nb_0.400	0.010	0.001	0.004	0.016	True
nb_0.425	0.011	0.002	0.006	0.019	True
nb_0.450	0.012	0.002	0.006	0.020	True
nb_0.475	0.014	0.002	0.007	0.023	True
nb_0.500	0.016	0.005	-0.003	0.038	False
nb_0.525	0.004	0.007	-0.023	0.031	False
nb_0.550	0.007	0.003	-0.004	0.019	False
nb_0.575	0.010	0.012	-0.041	0.058	False
nb_0.600	0.037	0.006	0.016	0.060	True
nb_0.625	0.033	0.005	0.011	0.055	True
nb_0.650	0.036	0.005	0.020	0.064	True
nb_0.675	0.030	0.005	0.010	0.051	True
nb_0.700	0.034	0.004	0.020	0.052	True
nb_0.725	0.037	0.004	0.024	0.056	True
nb_0.750	0.037	0.004	0.020	0.054	True
nb_0.775	0.043	0.005	0.025	0.064	True
nb_0.800	0.050	0.006	0.030	0.077	True

Table S25: The decision curve of Figure S3 as a table: BPIC14, the registered handover target, one-hot logistic regression, the clean register, the single temporal holdout, the intake block plus the free field as baseline. Every row is one operating point on RAW scores, and is therefore a score-threshold sensitivity analysis in the sense of Section 8.2 rather than a decision curve. The bands are the whole-family simultaneous max-t construction over this pair's declared decision-curve family, and 'sim resolved' says whether that band excludes zero. This family is NOT coverage-calibrated; Section 8.3 gives the counts under the empirical critical value beside these.

measure	description	median	min	max	largest first order median	n pairs first order exceeds	n pairs
equal-level	every level of every axis equally likely	0.560	0.233	0.793	0.287	6.000	19.000
reference	half the mass on the reference level of each axis	0.496	0.157	0.709	0.299	6.000	19.000
concentrated	eighty per cent on the reference level	0.317	0.065	0.638	0.392	12.000	19.000
envelope	200 Dirichlet draws of the level weights, primary pair	0.181	0.098	0.347	–	–	–

Table S26: The declared measures over specifications, and the total higher-order variance share under each. The magnitude moves with the measure; the qualitative conclusion, that no first-order index approaches the higher-order remainder, does not.

metric	unit	majority	one-number	uniform-beneficial	weighted-mean	majority (n suboptimal)	one-number (n suboptimal)	uniform-beneficial (n suboptimal)	weighted-mean (n suboptimal)
ap	average precision	0.000	0.002	0.027	0	1	3	17	0
auc	AUC	0.000	0.005	0.026	0	0	5	14	0
brier_skill	Brier skill	0.000	0.002	0.031	0	3	4	9	0
logloss_skill	log-loss skill	0.003	0.004	0.023	0	3	3	8	0
nagelkerke	Nagelkerke R2	0.025	0.042	0.037	0	6	6	7	0

Table S27: Four decision rules, INSIDE one instrument at a time. Mean excess regret across log–target pairs in that instrument's own units, and the number of pairs on which each rule is strictly suboptimal. No column averages across instruments, because there is no unit in which such an average would be expressed.

log	target	n train	n test	prev test	cal intercept	cal slope	brier	unseen
BPIC13_closed	handover	1040	447	0.275	-0.368	0.916	0.170	0.407
BPIC13_incidents	duration	5287	2267	0.154	-1.783	0.571	0.216	0.105
BPIC13_incidents	handover	5287	2267	0.616	-0.925	0.943	0.167	0.105
BPIC14	duration	32624	13982	0.427	-0.259	0.938	0.202	0.053
BPIC14	handover	32624	13982	0.945	0.378	0.994	0.037	0.053
BPIC15_1	duration	839	360	0.442	-1.204	0.997	0.228	0.753
BPIC15_1	handover	839	360	0.839	-0.061	1.255	0.066	0.753
BPIC15_3	duration	986	423	0.456	-0.746	0.928	0.214	0.095
BPIC15_3	handover	986	423	0.905	-0.649	1.437	0.069	0.095
BPIC15_4	duration	737	316	0.225	-1.927	0.029	0.351	0.649
BPIC15_4	handover	737	316	0.867	-1.349	0.302	0.119	0.649
BPIC15_5	duration	809	347	0.285	-0.793	0.290	0.244	0.867
BPIC17	duration	22056	9453	0.494	-0.060	0.758	0.244	0.060
BPIC19	duration	176213	75521	0.162	-3.163	0.278	0.352	0.299
Helpdesk	duration	3206	1374	0.451	-0.149	0.475	0.245	0.229
RoadFines	duration	105259	45111	0.429	-0.312	0.626	0.244	0.368
Sepsis	duration	735	315	0.444	-0.307	0.716	0.192	0.079
UCI498	duration	17442	7476	0.286	-0.619	0.811	0.176	0.192
UCI498	handover	17442	7476	0.420	0.017	1.230	0.071	0.192

Table S28: Calibration and unseen-category rates at the reference cell, one row per log–target pair in the same order as every other per-pair table. Every row is on the cohort and target the registered rules of Section 5 define, BPIC14's included; the case study's own cohort is smaller and its counts are in Section 7.1.

log	target	raw slope	platt slope	isotonic slope	raw ece	platt ece	isotonic ece
BPIC13_closed	handover	0.332	1.005	0.886	0.149	0.120	0.096
BPIC13_incidents	duration	0.575	1.052	0.902	0.295	0.304	0.296
BPIC13_incidents	handover	0.743	1.334	1.049	0.140	0.149	0.142
BPIC14	duration	0.945	1.117	1.050	0.056	0.056	0.056
BPIC14	handover	0.952	1.049	0.995	0.017	0.015	0.015
BPIC15_1	duration	0.731	1.812	1.118	0.208	0.172	0.152
BPIC15_1	handover	0.778	1.654	1.466	0.075	0.066	0.050
BPIC15_3	duration	0.886	1.225	1.109	0.190	0.152	0.152
BPIC15_3	handover	0.450	1.247	0.904	0.071	0.057	0.039
BPIC15_4	duration	0.025	0.028	-0.013	0.364	0.313	0.340
BPIC15_4	handover	0.282	0.701	0.456	0.086	0.071	0.080
BPIC15_5	duration	0.263	0.853	0.414	0.252	0.209	0.211
BPIC17	duration	0.868	1.165	1.031	0.023	0.019	0.017
BPIC19	duration	0.293	0.355	0.372	0.339	0.338	0.338
Helpdesk	duration	0.490	0.731	0.622	0.088	0.066	0.065
RoadFines	duration	0.688	0.720	1.377	0.081	0.086	0.081
Sepsis	duration	0.456	0.830	0.554	0.147	0.086	0.089
UCI498	duration	0.882	1.534	1.333	0.104	0.185	0.175
UCI498	handover	1.077	1.453	1.203	0.041	0.063	0.047

Table S29: Calibration before and after a training-only recalibration, one row per log–target pair: the median calibration slope and expected calibration error over the arms of the baseline ladder, raw and under each calibrator. A slope far from one is a score whose thresholds are not cost ratios; only curves computed on calibrated probabilities carry a probability-threshold reading.

evidence	value	establishes
interactions in the file	147004.000	the population the knowledge reference is written in
interactions that never became incidents	94250.000	the field is written by the service-desk process, not by the incident process
knowledge reference populated on those	1.000	the same, quantitatively
interactions CLOSED before the incident was opened	5.000	airtight pre-incident availability, on five cases, which is not enough to rest a headline on
interactions whose recorded handling time had elapsed before the incident was opened	41413.000	that the desk had worked the interaction, which is not a timestamped field-value history
knowledge reference populated on the joined cohort	0.911	coverage, not timing
base AUC from the knowledge reference alone	0.648	the field is highly predictive, which is a reason for suspicion and not for confidence

Table S30: What the interaction file establishes, and what it does not. Case study cohort, reassignment target.

mechanism	level	populated	V	spread over seeds	seeds
clean	1.000	1.000	0.103	0.000	1
corrupt	0.050	1.000	0.099	0.003	3
corrupt	0.150	1.000	0.086	0.001	3
corrupt	0.300	1.000	0.069	0.002	3
duplicate	0.150	1.000	0.103	0.001	3
duplicate	0.500	1.000	0.104	0.001	3
duplicate	1.000	1.000	0.102	0.001	3
mask_common	0.500	0.521	0.071	0.000	1
mask_random	0.500	0.502	0.068	0.004	3
mask_rare	0.250	0.270	0.041	0.000	1
mask_rare	0.500	0.505	0.082	0.000	1
mask_rare	0.750	0.762	0.096	0.000	1
stale	1.000	0.984	0.103	0.000	1
stale_lag	0.100	0.982	0.107	0.000	1
stale_lag	0.250	0.972	0.106	0.000	1
stale_lag	0.500	0.940	0.107	0.000	1

Table S31: Register-quality mechanisms on the case study's cohort (45,455 incidents) and reassignment target, at the reference cell. 'level' is the share populated for the population mechanisms, the share corrupted for accuracy, the share of identities split for reconciliation and the lag for discovery. Stochastic mechanisms are averaged over 20 seeds and the spread across seeds is printed beside the mean, because mechanism variability and sampling variability are different quantities. The corpus-cohort version of this table is in Supplement S15.

mechanism	levels	seeds	integral	coarse	shift	seed s.d.	V complete	V empty
content	11	3	0.049	0.047	0.001	0.002	0.085	-0.001
mask_rare	11	1	0.069	0.066	0.003	0.000	0.103	-0.003
propensity	11	3	0.056	0.054	0.002	0.002	0.086	-0.001
time	11	1	0.077	0.088	0.011	0.000	0.115	-0.001

Table S32: Register-quality mechanisms as severity CURVES rather than points, on the case study's cohort and reassignment target. The integral is taken against a uniform measure on severity over 11 levels; 'grid shift' is how much coarsening that grid to five levels moves it, which is the check that the summary is a property of the curve and not of the grid.

b lo	b hi	V lo	V hi	D	D lo	D hi	R	R fieller lo	R fieller hi	fieller kind
B_half	B_intake	0.1828	0.1838	-0.0010	-0.0049	0.0033	-0.0056	-0.0270	0.0156	bounded
B_half	B_intake_g	0.1828	0.1034	0.0793	0.0619	0.0896	0.4341	0.3572	0.5127	bounded
B_half	B_intake_g_km	0.1828	-0.0029	0.1857	0.1759	0.2025	1.0157	0.9904	1.0425	bounded
B_intake	B_intake_g	0.1838	0.1034	0.0804	0.0626	0.0914	0.4372	0.3611	0.5151	bounded
B_intake	B_intake_g_km	0.1838	-0.0029	0.1867	0.1773	0.2040	1.0156	0.9905	1.0422	bounded
B_intake_g	B_intake_g_km	0.1034	-0.0029	0.1063	0.0965	0.1308	1.0277	0.9831	1.0771	bounded

Table S33: Absolute absorption $D = V(f \mid B_0) - V(f \mid B_1)$ first, with its own interval, then the ratio R with its Fieller set and the set's kind, on the case study's cohort and reassignment target, in ROC AUC. 0 of 30 reductions across all five instruments have a set that is not a bounded interval; for those a percentile interval is not a summary of anything.

mechanism	level	populated	V
clean	1.000	1.000	0.167
mask_rare	0.750	0.760	0.140
mask_rare	0.500	0.515	0.104
mask_rare	0.250	0.268	0.037
mask_random	0.500	0.502	0.095
mask_common	0.500	0.512	0.106
corrupt	0.050	1.000	0.153
corrupt	0.150	1.000	0.132
corrupt	0.300	1.000	0.097
duplicate	0.150	1.000	0.167
duplicate	0.500	1.000	0.165
duplicate	1.000	1.000	0.163
stale	1.000	0.984	0.167
stale_lag	0.100	0.982	0.169
stale_lag	0.250	0.971	0.169
stale_lag	0.500	0.941	0.164

Table S34: Register-quality mechanisms on the REGISTERED cohort (46,606 cases) and the handover target, at the reference cell, including a discovery lag and a sweep of the two accuracy mechanisms. The case study's own cohort and target are in Table S31.

mechanism	operational failure mode	n levels	n seeds	integral	grid shift	seed s.d.	V complete	V empty
content	discovery: coverage depends on the item's type	11	20	0.090	0.015	0.031	0.146	0.000
corrupt	accuracy: a share of rows carry the wrong identity	11	20	0.113	0.010	0.005	0.068	0.167
duplicate	reconciliation: one item exists under two keys	11	20	0.165	0.001	0.001	0.163	0.167
mask_common	population: the core identities are absent	11	1	0.104	0.018	0.000	0.167	0.014
mask_random	population: absent independently of everything	11	20	0.088	0.016	0.004	0.167	0.000
mask_rare	population: the long tail of identities is absent	11	1	0.092	0.016	0.000	0.167	0.014
propensity	missing where the work is unusual: a training-estimated propensity drives the missingness	11	20	0.069	0.010	0.004	0.119	0.000
time	discovery: the register was populated late	11	1	0.105	0.005	0.000	0.167	0.000

Table S35: Register-quality mechanisms as severity CURVES rather than points, on the primary log at the reference cell. Each row gives the operational failure mode the mechanism represents, the number of severities and seeds, the integral of the increment against a uniform measure on severity, how much coarsening the severity grid by a factor of four moves that integral, the spread across seeds, and the endpoints.

metric	b lo	b hi	V lo	V hi	D	D lo	D hi	R	R fieller lo	R fieller hi	fieller kind
ap	B_half	B_intake	0.031	0.030	0.000	0.000	0.002	0.014	-0.027	0.052	bounded
ap	B_half	B_intake_g	0.031	0.019	0.012	0.009	0.016	0.390	0.276	0.498	bounded
ap	B_half	B_intake_g_km	0.031	0.001	0.030	0.026	0.033	0.976	0.949	1.004	bounded
ap	B_intake	B_intake_g	0.030	0.019	0.012	0.008	0.016	0.381	0.266	0.492	bounded
ap	B_intake	B_intake_g_km	0.030	0.001	0.029	0.026	0.033	0.976	0.948	1.004	bounded
ap	B_intake_g	B_intake_g_km	0.019	0.001	0.018	0.014	0.021	0.961	0.917	1.007	bounded
auc	B_half	B_intake	0.260	0.257	0.003	-0.002	0.015	0.010	-0.021	0.039	bounded
auc	B_half	B_intake_g	0.260	0.167	0.092	0.064	0.127	0.356	0.239	0.469	bounded
auc	B_half	B_intake_g_km	0.260	0.008	0.252	0.229	0.272	0.970	0.946	0.995	bounded
auc	B_intake	B_intake_g	0.257	0.167	0.090	0.063	0.126	0.350	0.229	0.466	bounded
auc	B_intake	B_intake_g_km	0.257	0.008	0.249	0.228	0.271	0.970	0.945	0.995	bounded
auc	B_intake_g	B_intake_g_km	0.167	0.008	0.160	0.122	0.179	0.954	0.917	0.992	bounded
brier_skill	B_half	B_intake	0.269	0.269	0.000	-0.003	0.003	0.001	-0.009	0.012	bounded
brier_skill	B_half	B_intake_g	0.269	0.247	0.022	-0.006	0.049	0.081	-0.026	0.177	bounded
brier_skill	B_half	B_intake_g_km	0.269	0.004	0.265	0.229	0.290	0.985	0.943	1.028	bounded
brier_skill	B_intake	B_intake_g	0.269	0.247	0.021	-0.004	0.049	0.080	-0.029	0.177	bounded
brier_skill	B_intake	B_intake_g_km	0.269	0.004	0.265	0.227	0.292	0.985	0.943	1.028	bounded
brier_skill	B_intake_g	B_intake_g_km	0.247	0.004	0.243	0.208	0.274	0.983	0.938	1.031	bounded
logloss_skill	B_half	B_intake	0.321	0.320	0.001	-0.002	0.002	0.003	-0.004	0.010	bounded
logloss_skill	B_half	B_intake_g	0.321	0.241	0.080	0.060	0.102	0.249	0.188	0.308	bounded
logloss_skill	B_half	B_intake_g_km	0.321	0.015	0.306	0.279	0.328	0.954	0.924	0.985	bounded
logloss_skill	B_intake	B_intake_g	0.320	0.241	0.079	0.060	0.101	0.247	0.186	0.306	bounded
logloss_skill	B_intake	B_intake_g_km	0.320	0.015	0.305	0.279	0.329	0.954	0.924	0.985	bounded
logloss_skill	B_intake_g	B_intake_g_km	0.241	0.015	0.226	0.203	0.244	0.939	0.900	0.980	bounded
nagelkerke	B_half	B_intake	0.363	0.362	0.001	-0.003	0.002	0.003	-0.005	0.010	bounded
nagelkerke	B_half	B_intake_g	0.363	0.266	0.096	0.073	0.121	0.266	0.203	0.326	bounded
nagelkerke	B_half	B_intake_g_km	0.363	0.015	0.348	0.318	0.372	0.959	0.932	0.987	bounded
nagelkerke	B_intake	B_intake_g	0.362	0.266	0.096	0.074	0.121	0.264	0.201	0.324	bounded
nagelkerke	B_intake	B_intake_g_km	0.362	0.015	0.347	0.318	0.372	0.959	0.932	0.987	bounded
nagelkerke	B_intake_g	B_intake_g_km	0.266	0.015	0.252	0.226	0.271	0.944	0.908	0.982	bounded

Table S36: Absolute absorption D first, the ratio R second, with its Fieller set and the set's kind, on the REGISTERED cohort and the handover target, over all five instruments. The case study's own cohort and target are in Table S33.

world	bias vs limit	width (basic)	fixed pct	fixed basic	fixed bc	nested pct	nested basic	nested bc	m-out-of-n
drift	0.000	0.037	0.919	0.917	0.914	0.966	0.919	0.920	–
imbalanced	-0.010	0.116	0.906	0.906	0.900	0.933	0.933	0.923	–
linear	-0.002	0.060	0.925	0.931	0.915	0.939	0.931	0.921	0.768
noisy	-0.002	0.036	0.926	0.924	0.916	0.936	0.948	0.939	–
nonlinear	-0.002	0.044	0.923	0.923	0.924	0.940	0.929	0.929	0.758
sparse	-0.012	0.034	0.631	0.643	0.620	0.372	0.899	0.631	0.728

Table S37: Coverage of the estimator's own limit V_{limit} in each simulated world, under seven interval constructions, at the replicate count of Section 10, with the estimator's bias against that limit and the mean width of the interval this paper reports. Nominal is 0.95; the Monte Carlo standard error on a coverage is 0.7 percentage points, so differences below about a point are not differences. An empty cell is a construction not priced on that world.

n train	K	basic coverage	pct coverage	basic mean width	pct mean width	replicates	coverage se (pp)
2000	20	0.887	0.887	0.085	0.085	150	2.588
2000	200	0.780	0.480	0.042	0.042	150	3.382
2000	1000	0.453	0.100	0.064	0.064	150	4.065
4000	20	0.900	0.907	0.057	0.057	150	2.449
4000	200	0.907	0.500	0.031	0.031	150	2.375
4000	1000	0.507	0.080	0.047	0.047	150	4.082
8000	20	0.893	0.900	0.039	0.039	150	2.520
8000	200	0.893	0.480	0.021	0.021	150	2.520
8000	1000	0.613	0.033	0.034	0.034	150	3.976
8000	3019	0.213	0.000	0.026	0.026	150	3.345

Table S38: Sample size and register cardinality varied jointly. Coverage of V_{limit} and mean interval width for the nested percentile and the nested basic construction. The last row is the cardinality regime of the case study's own register. *How well these are determined.* Every cell is 150 replicates, not the 1,000 of the core experiment, and the coverages here are far from nominal — so the Monte Carlo standard error of a coverage in this table has a median of 3.0 percentage points and reaches 4.1, against the 0.7 that Section 10.3 quotes for the core. It is 3.7 times larger, it is computed at each cell's own observed rate rather than at the nominal one, and it is a column here rather than a sentence because the differences down this table are read row against row.

world	block multiplier	nested basic	nested pct
linear	0.500	0.896	0.944
linear	1.000	0.920	0.908
linear	2.000	0.916	0.932
sparse	0.500	0.856	0.392
sparse	1.000	0.836	0.332
sparse	2.000	0.872	0.408

Table S39: Three block lengths around $n^{1/3}$: the multiplier column is the factor applied to the rule of thumb, so 0.5 halves the block and 2 doubles it. Coverage of V_{limit}.

family	draws are	K	draws	multiplier	mult., MC upper	empirical	emp., OS upper	Rademacher	per-cell
decision-curve	degenerate	534	40	0.392	0.394	0.877	0.889	0.281	0.902
decision-curve	degenerate	1100	80	0.579	0.687	0.892	0.908	0.244	0.922
decision-curve	degenerate	2966	150	0.853	0.854	0.902	0.920	0.821	0.932
decision-curve	gaussian	534	40	0.748	0.756	0.549	0.726	0.601	0.905
decision-curve	gaussian	1100	80	0.859	0.866	0.786	0.923	0.805	0.928
decision-curve	gaussian	2966	150	0.887	0.891	0.846	0.919	0.853	0.937
decision-curve	heavy	534	40	0.760	0.766	0.840	0.894	0.631	0.905
decision-curve	heavy	1100	80	0.838	0.842	0.899	0.920	0.790	0.929
decision-curve	heavy	2966	150	0.854	0.855	0.907	0.923	0.823	0.939
whole-surface	degenerate	120	33	0.835	0.842	0.759	0.888	0.771	0.902
whole-surface	degenerate	180	80	0.927	0.931	0.887	0.974	0.900	0.927
whole-surface	degenerate	480	150	0.936	0.942	0.913	0.969	0.926	0.937
whole-surface	gaussian	120	33	0.824	0.827	0.710	0.821	0.744	0.898
whole-surface	gaussian	180	80	0.900	0.903	0.858	0.955	0.876	0.927
whole-surface	gaussian	480	150	0.909	0.914	0.887	0.948	0.895	0.937
whole-surface	heavy	120	33	0.822	0.827	0.740	0.865	0.749	0.900
whole-surface	heavy	180	80	0.909	0.910	0.878	0.966	0.888	0.924
whole-surface	heavy	480	150	0.928	0.929	0.902	0.967	0.917	0.935

Table S40: Family-wise coverage of the max-t band against a known answer, over synthetic families matched to this corpus in size, draw count, cross-cell correlation and per-cell excess kurtosis. Nominal is 0.95; the five middle columns are the five candidate critical values and the last is the control. *The bootstrap here is ideal* — the draws come from the sampling distribution itself — so every coverage in this table is an upper bound on what the real procedure attains, and the control says why: *per-cell* is the basic interval the band is built from, on the same draws, and it is short of nominal before any multiplicity correction is applied. *Regimes.* `gaussian` is the case the multiplier bootstrap is derived for; `heavy` carries this corpus's measured tails; `degenerate` adds the cells whose net-benefit difference is zero by arithmetic, which the corpus's decision-curve families contain and its surface families do not. Every cell is 2,000 replicates, so no coverage here is determined worse than 1.1%.

world	n train	K	K / n	replicates	coverage at c = 1	se	c	c lo	c hi	c applied	c log-linear
linear	180000	200	0.001	500	0.882	0.014	1.352	1.174	1.482	1.177	1.000
linear	8000	20	0.003	500	0.916	0.012	1.154	1.073	1.307	1.177	1.000
linear	4000	20	0.005	500	0.898	0.014	1.219	1.101	1.313	1.177	1.056
linear	2000	20	0.010	500	0.924	0.012	1.108	1.024	1.228	1.177	1.134
linear	50000	625	0.013	498	0.902	0.013	1.296	1.135	1.689	1.177	1.160
linear	8000	100	0.013	500	0.936	0.011	1.096	0.960	1.203	1.177	1.160
sparse	8000	100	0.013	500	0.892	0.014	1.317	1.220	1.453	1.177	1.160
linear	2000	25	0.013	500	0.908	0.013	1.188	1.072	1.277	1.177	1.160
linear	750	10	0.013	500	0.920	0.012	1.117	1.029	1.207	1.177	1.168
linear	4000	100	0.025	500	0.896	0.014	1.179	1.087	1.296	1.177	1.245
linear	8000	200	0.025	499	0.932	0.011	1.093	0.985	1.264	1.177	1.245
sparse	4000	200	0.050	500	0.884	0.014	1.182	1.131	1.299	1.177	1.337
linear	2000	100	0.050	500	0.912	0.013	1.234	1.055	1.326	1.177	1.337
linear	4000	200	0.050	500	0.888	0.014	1.195	1.119	1.337	1.177	1.337
linear	8000	400	0.050	500	0.884	0.014	1.152	1.104	1.226	1.177	1.337
linear	750	38	0.051	500	0.926	0.012	1.103	0.989	1.259	1.177	1.338
sparse	2000	200	0.100	500	0.836	0.017	1.333	1.224	1.372	1.310	1.435
linear	500	50	0.100	500	0.874	0.015	1.250	1.141	1.460	1.310	1.435
linear	4000	400	0.100	499	0.796	0.018	1.287	1.221	1.399	1.310	1.435
linear	2000	200	0.100	500	0.844	0.016	1.316	1.253	1.417	1.310	1.435
linear	8000	1000	0.125	496	0.655	0.021	1.526	1.417	1.600	1.441	1.468
linear	2000	250	0.125	500	0.792	0.018	1.404	1.284	1.485	1.441	1.468
linear	750	94	0.125	500	0.872	0.015	1.310	1.154	1.396	1.441	1.468
linear	2000	400	0.200	500	0.758	0.019	1.562	1.398	1.670	1.562	1.540
sparse	4000	1000	0.250	498	0.460	0.022	1.716	1.672	1.849	1.727	1.575
linear	4000	1000	0.250	500	0.520	0.022	1.739	1.632	1.805	1.727	1.575
linear	8000	3019	0.377	488	0.305	0.021	1.881	1.804	1.988	1.791	1.643
linear	500	200	0.400	500	0.760	0.019	1.450	1.353	1.656	1.791	1.653
linear	2000	1000	0.500	500	0.462	0.022	1.869	1.788	2.024	1.869	1.691

Table S41: The coverage calibration. For each cell of the (n, K) plane: the coverage the reported interval actually attains, and the factor c by which it must be widened to attain the nominal level, with the Monte Carlo interval of that factor. c is the 0.95 quantile of the studentised error and is computed, not searched for. 'c applied' is the monotone regression the corpus is calibrated with; 'c log-linear' is the parametric fit whose slope Section 6.3 quotes, printed beside it because it sits below the measured factor at small ratios and would under-correct there.

References

[1] van der Aalst, W.M.P., 2016. Process Mining: Data Science in Action. 2nd ed., Springer. doi:10.1007/978-3-662-49851-4.

[2] Bouthillier, X., Delaunay, P., Bronzi, M., et al., 2021. Accounting for variance in machine learning benchmarks, in: Proceedings of Machine Learning and Systems (MLSys).

[3] Breznau, N., Rinke, E.M., Wuttke, A., et al., 2022. Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty. Proceedings of the National Academy of Sciences 119, e2203150119. doi:10.1073/pnas.2203150119.

[4] Çarka, J., Esposito, M., Falessi, D., 2022. On effort-aware metrics for defect prediction. Empirical Software Engineering 27, 152. doi:10.1007/s10664-022-10186-7.

[5] Chernozhukov, V., Chetverikov, D., Kato, K., 2013. Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. The Annals of Statistics 41, 2786–2819. doi:10.1214/13-AOS1161.

[6] Comuzzi, M., Kim, S., Ko, J., Salamov, M., Cappiello, C., Pernici, B., 2025. On the impact of low-quality activity labels in predictive process monitoring, in: Business Process Management Workshops. doi:10.1007/978-3-031-82225-4_15.

[7] Cook, N.R., 2007. Use and misuse of the receiver operating characteristic curve in risk prediction. Circulation 115, 928–935. doi:10.1161/CIRCULATIONAHA.106.672402.

[8] Covert, I., Lundberg, S., Lee, S.I., 2020. Understanding global feature contributions with additive importance measures, in: Advances in Neural Information Processing Systems 33.

[9] D'Amour, A., Heller, K., Moldovan, D., et al., 2022. Underspecification presents challenges for credibility in modern machine learning. Journal of Machine Learning Research 23, 1–61.

[10] Davison, A.C., Hinkley, D.V., 1997. Bootstrap Methods and Their Application. Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, Cambridge. doi:10.1017/CBO9780511802843.

[11] Fisher, A., Rudin, C., Dominici, F., 2019. All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. Journal of Machine Learning Research 20, 1–81.

[12] Ghorbani, A., Zou, J., 2019. Data Shapley: Equitable valuation of data for machine learning, in: Proceedings of the 36th International Conference on Machine Learning, pp. 2242–2251.

[13] Jia, R., Dao, D., Wang, B., Hubis, F.A., Hynes, N., Gürel, N.M., Li, B., Zhang, C., Song, D., Spanos, C.J., 2019. Towards efficient data valuation based on the Shapley value, in: Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics, pp. 1167–1176.

[14] Kapoor, S., Narayanan, A., 2023. Leakage and the reproducibility crisis in machine-learning-based science. Patterns 4, 100804. doi:10.1016/j.patter.2023.100804.

[15] Kaufman, S., Rosset, S., Perlich, C., Stitelman, O., 2012. Leakage in data mining: Formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data 6, 15:1–15:21. doi:10.1145/2382577.2382579.

[16] Kerr, K.F., McClelland, R.L., Brown, E.R., Lumley, T., 2011. Evaluating the incremental value of new biomarkers with integrated discrimination improvement. American Journal of Epidemiology 174, 364–374. doi:10.1093/aje/kwr086.

[17] Künsch, H.R., 1989. The jackknife and the bootstrap for general stationary observations. The Annals of Statistics 17, 1217–1241. doi:10.1214/aos/1176347265.

[18] Liu, C., Liu, W., Guo, N., Song, R., Gu, Y., Cheng, L., 2025. Comparative evaluation of encoding techniques for workflow process remaining time prediction for cloud applications. Journal of Cloud Computing 14. doi:10.1186/s13677-025-00763-8.

[19] Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems 30, pp. 4765–4774.

[20] Marrone, M., Kolbe, L.M., 2011. Impact of IT service management frameworks on the IT organization. Business & Information Systems Engineering 3, 5–18. doi:10.1007/s12599-010-0141-5.

[21] Marx, C.T., Calmon, F.d.P., Ustun, B., 2020. Predictive multiplicity in classification, in: Proceedings of the 37th International Conference on Machine Learning, pp. 6765–6774.

[22] Micci-Barreca, D., 2001. A preprocessing scheme for high-cardinality categorical attributes in classification and prediction problems. ACM SIGKDD Explorations Newsletter 3, 27–32. doi:10.1145/507533.507538.

[23] Mohammed, S., Budach, L., Feuerpfeil, M., Ihde, N., Nathansen, A., Noack, N., Patzlaff, H., Naumann, F., Harmouch, H., 2025. The effects of data quality on machine learning performance on tabular data. Information Systems 132, 102549. doi:10.1016/j.is.2025.102549.

[24] Patel, C.J., Burford, B., Ioannidis, J.P.A., 2015. Assessment of vibration of effects due to model specification can demonstrate the instability of observational associations. Journal of Clinical Epidemiology 68, 1046–1058. doi:10.1016/j.jclinepi.2015.05.029.

[25] Pencina, M.J., D'Agostino Sr., R.B., D'Agostino Jr., R.B., Vasan, R.S., 2008. Evaluating the added predictive ability of a new marker: From area under the ROC curve to reclassification and beyond. Statistics in Medicine 27, 157–172. doi:10.1002/sim.2929.

[26] Rama-Maneiro, E., Vidal, J.C., Lama, M., 2023. Deep learning for predictive business process monitoring: Review and benchmark. IEEE Transactions on Services Computing 16, 739–756. doi:10.1109/TSC.2021.3139807.

[27] Semenova, L., Rudin, C., Parr, R., 2022. On the existence of simpler machine learning models. Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency , 1827–1858doi:10.1145/3531146.3533232.

[28] Senderovich, A., Di Francescomarino, C., Ghidini, C., Jorbina, K., Maggi, F.M., 2018. Intra and inter-case features in predictive process monitoring. Information Systems 74, 102–113. doi:10.1016/j.is.2017.10.005.

[29] Silberzahn, R., Uhlmann, E.L., Martin, D.P., et al., 2018. Many analysts, one data set: Making transparent how variations in analytic choices affect results. Advances in Methods and Practices in Psychological Science 1, 337–356. doi:10.1177/2515245917747646.

[30] Simonsohn, U., Simmons, J.P., Nelson, L.D., 2020. Specification curve analysis. Nature Human Behaviour 4, 1208–1214. doi:10.1038/s41562-020-0912-z.

[31] Steegen, S., Tuerlinckx, F., Gelman, A., Vanpaemel, W., 2016. Increasing transparency through a multiverse analysis. Perspectives on Psychological Science 11, 702–712. doi:10.1177/1745691616658637.

[32] Strobl, C., Boulesteix, A.L., Kneib, T., Augustin, T., Zeileis, A., 2008. Conditional variable importance for random forests. BMC Bioinformatics 9, 307. doi:10.1186/1471-2105-9-307.

[33] Suriadi, S., Andrews, R., ter Hofstede, A.H.M., Wynn, M.T., 2017. Event log imperfection patterns for process mining. Information Systems 64, 132–150. doi:10.1016/j.is.2016.07.011.

[34] Teinemaa, I., Dumas, M., La Rosa, M., Maggi, F.M., 2019. Outcome-oriented predictive process monitoring: Review and benchmark. ACM Transactions on Knowledge Discovery from Data 13, 17:1–17:57. doi:10.1145/3301300.

[35] Van Calster, B., McLernon, D.J., van Smeden, M., Wynants, L., Steyerberg, E.W., 2019. Calibration: The Achilles heel of predictive analytics. BMC Medicine 17, 230. doi:10.1186/s12916-019-1466-7.

[36] VanderWeele, T.J., Ding, P., 2017. Sensitivity analysis in observational research: Introducing the e-value. Annals of Internal Medicine 167, 268–274. doi:10.7326/M16-2607.

[37] Verenich, I., Dumas, M., La Rosa, M., Maggi, F.M., Teinemaa, I., 2019. Survey and cross-benchmark comparison of remaining time prediction

methods in business process monitoring. ACM Transactions on Intelligent Systems and Technology 10, 34:1–34:34. doi:10.1145/3331449.

[38] Vickers, A.J., 2008. Decision analysis for the evaluation of diagnostic tests, prediction models, and molecular markers. The American Statistician 62, 314–320. doi:10.1198/000313008X370302.

[39] Vickers, A.J., van Calster, B., Steyerberg, E.W., 2019. A simple, step-by-step guide to interpreting decision curve analysis. Diagnostic and Prognostic Research 3, 18. doi:10.1186/s41512-019-0064-7.

[40] Vickers, A.J., Elkin, E.B., 2006. Decision curve analysis: A novel method for evaluating prediction models. Medical Decision Making 26, 565–574. doi:10.1177/0272989X06295361.

[41] Vickers, A.J., Van Calster, B., Steyerberg, E.W., 2016. Net benefit approaches to the evaluation of prediction models, molecular markers, and diagnostic tests. BMJ 352, i6. doi:10.1136/bmj.i6.

[42] Weytjens, H., De Weerdt, J., 2022. Creating unbiased public benchmark datasets with data leakage prevention for predictive process monitoring, in: Business Process Management Workshops, pp. 18–29. doi:10.1007/978-3-030-94343-1_2.

[43] Williamson, B.D., Gilbert, P.B., Carone, M., Simon, N., 2021. Nonparametric variable importance assessment using machine learning techniques. Biometrics 77, 9–22. doi:10.1111/biom.13392.